Early-Life Disadvantage and Later-Life Mental and Cognitive Health in India

Evidence from the Longitudinal Ageing Study in India (LASI Wave 1)

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Abstract

Do the hardships of a poor childhood in India—growing up without money, falling gravely ill, spending weeks bedridden—still echo in the mind half a century later? Drawing on Wave 1 of the Longitudinal Ageing Study in India (66,606 adults aged 45 and older), we followed early-life disadvantage into later-life depression and cognition using survey-weighted modified Poisson regression. The two outcomes told opposite stories. Depression carried a lasting scar: poor childhood health, childhood poverty, and prolonged bedridden illness each raised the prevalence of late-life depressive symptoms by roughly 16–21% (up to 42% under the strictest definition of childhood ill-health), and neither adult education nor adult wealth washed this away. Cognition, by contrast, healed: once adult schooling and household consumption were taken into account, the childhood-poverty gap in memory and orientation disappeared entirely (APR = 0.93). Climbing out of poverty in adulthood rebuilt the mind but not the mood. The policy implication is direct—healthy ageing begins in childhood, and investments in early nutrition, universal schooling, and community mental-health care are investments in the India of fifty years from now.

Keywords: Life-Course Epidemiology, Cognitive Ageing, Late-Life Depression, LASI Wave 1, Survey-Weighted Poisson Regression, Social Determinants of Health, India.

Statement of Contributions

This report is the product of a shared effort in which each member owned a distinct phase of the research pipeline, from the opening question to the final edit. **Ankit Kumar Jha** coordinated the project, shaped the research design and central research question, and integrated the separate strands into a single coherent study. **Harshit Tripathi** handled data acquisition, cleaning, variable construction, and the statistical modelling underlying every table. **Shivam Sah** led the literature review and background research and assembled the references that anchor the study in the sociological canon. **Ankit Kholwad** built the figures, charts, and visual narrative that carry the argument. **Neelatmajam Dwivedi** prepared and edited the written report and developed the accompanying presentation. All members reviewed the final manuscript and endorse its conclusions.

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1 Introduction and Theoretical Framework

India is ageing before it has finished growing rich. The men and women at the centre of this report were born into a country of famine, fever, and empty schoolhouses—and have lived to see a different nation entirely. Our question is deceptively simple: how much of that lost childhood did they carry into old age?

1.1 The Demographic Imperative: Population Ageing in India

India is currently undergoing one of the most rapid and expansive demographic transitions in global history. Driven by dramatic reductions in infant mortality and fertility alongside substantial gains in average life expectancy, the proportion and absolute count of older individuals are expanding at unprecedented rates. As of the 2021 Census projections, India was home to approximately 140 million adults aged 60 years and older. By the year 2050, this demographic cohort is projected to surge beyond 340 million, constituting over 20% of the national population.

This demographic transition in India differs fundamentally from the historical ageing experiences of Western Europe, North America, and East Asia. High-income nations experienced population ageing after achieving widespread industrialization, high literacy rates, and institutionalized social welfare and universal healthcare systems. In contrast, India is "ageing before getting rich." The current generation of older adults in India was born between the 1930s and 1970s—a period characterized by acute rural poverty, recurring food insecurity, high rates of childhood infectious illnesses, and minimal access to formal primary education.

As this large cohort reaches older ages, two interconnected public health challenges have emerged as leading drivers of disability and loss of functional independence: **cognitive impairment and dementia**, and **late-life depression**. While cognitive decline diminishes an older person's autonomy, memory, and executive capacity, depression severely impairs daily functioning, exacerbates chronic somatic conditions, and elevates mortality. Understanding whether these debilitating conditions stem primarily from late-life lifestyle factors and chronic physical illnesses, or whether their seeds were planted decades earlier in childhood, is essential for designing effective, equitable public health interventions across the Indian life course.

1.2 Life-Course Epidemiological Paradigms

To investigate how early-life experiences influence health outcomes in old age, researchers rely on **life-course epidemiology**—the study of long-term biological, behavioral, and psychosocial pathways that link physical and social exposures during gestation, childhood, and adolescence to adult health. Gerontologists and epidemiologists conceptualize these connections through three primary, non-mutually exclusive theoretical paradigms:

(a) **The Sensitive or Critical Period Model (Biological Vulnerability):** This paradigm posits that adverse shocks during critical windows of anatomical and physiological development—such as intrauterine life, infancy, and early childhood—leave indelible biological marks on the body. Prolonged undernutrition, severe infectious diseases, or extreme psychological stress during these early windows can permanently alter neurodevelopmental structures, program the hypothalamic-pituitary-adrenal (HPA) stress axis toward chronic

hyper-reactivity, and establish persistent, low-grade systemic inflammation. According to this model, these biological vulnerabilities persist across the entire lifespan, predisposing the individual to elevated risks of depression, cerebrovascular disease, and functional decline in older adulthood, regardless of subsequent upward socioeconomic mobility achieved in mid-life.

(b) **The Accumulation of Disadvantage Model (Cumulative Chains of Risk):** Rather than focusing solely on a single critical window, this model suggests that health damages accumulate incrementally over time. Being born into an impoverished household initiates a sequential chain of adverse exposures: limited family financial resources lead to inadequate nutrition and early school dropout; low educational attainment channels individuals into hazardous, low-wage manual labor; poor adult working and living conditions lead to chronic environmental exposures, lack of healthcare access, and early onset of chronic morbidities such as hypertension and diabetes. Across 50 to 70 years of life, these cumulative insults gradually erode cognitive vitality and mental resilience.

(c) **The Pathway or Social Mobility Model (Adult Attenuation):** This model argues that early-life disadvantage influences later-life health primarily because it determines adult socioeconomic attainment. In this view, early childhood poverty is not necessarily a direct biological destiny; rather, its primary effect is that it restricts access to formal schooling and economic capital in adulthood. If adult education, occupation, and household wealth act as key mediators along this pathway, then statistically controlling for adult socioeconomic attainment should substantially attenuate (or completely eliminate) the observed statistical association between childhood disadvantage and later-life health.

1.3 Study Hypotheses

To systematically evaluate these theoretical models within the Indian context, we formulate five foundational hypotheses grounded in life-course theory:

Hypothesis 1: Childhood Socioeconomic Deprivation: Individuals who grew up in impoverished or financially average households will exhibit a higher prevalence of cognitive impairment and depressive symptoms in later life compared to those from well-off childhood backgrounds.

Hypothesis 2: Poor Childhood Health: Individuals who experienced poor or fair general health before age 16 will display significantly elevated rates of cognitive impairment and depressive symptoms in older age relative to peers who enjoyed good childhood health.

Hypothesis 3: Severe Childhood Bedridden Illness: Having experienced a severe illness that caused the respondent to be bedridden for one month or longer during childhood will independently elevate the prevalence of later-life cognitive impairment and depressive symptoms.

Hypothesis 4: Age Heterogeneity (Divergence vs. Selection): The strength of association between early disadvantage and late-life outcomes will vary systematically across

stages of ageing (mid-life: 45–59 years; young-old: 60–74 years; oldest-old: 75+ years), reflecting either cumulative divergence (widening inequalities) or mortality selection (survivor bias narrowing inequalities).

Hypothesis 5: **Adult Socioeconomic Attenuation:** The crude statistical associations between childhood socioeconomic status and late-life cognitive and mental health outcomes will be substantially attenuated upon sequential adjustment for adult educational attainment and household per-capita consumption expenditure.

1.4 Methodological Challenges and Analytical Framework

Investigating life-course health disparities in low- and middle-income countries presents unique methodological challenges that standard epidemiological models must carefully accommodate:

1. **Cognitive Assessment in High-Illiteracy Populations:** In standard cognitive batteries, items testing mental arithmetic (such as serial 7 subtractions and backward counting) or written tasks cannot be administered to respondents without formal education. When these items are skipped and scored as zero points, the resulting test measures formal schooling rather than true neurological vitality. Our study addresses this by establishing an education-fair primary cognitive outcome focused on universally administered verbal memory and orientation tasks, while evaluating broader multi-domain batteries in sensitivity analyses.

2. **Multistage Complex Survey Sampling:** Older adults are nested within villages, urban wards, and states. Standard regression models that ignore clustering and stratification underestimate standard errors. We apply design-based Taylor series linearization across 72 geographical strata and 2,428 sampling clusters.

3. **Common Complete-Case Samples:** To ensure that changes in effect estimates across sequential models reflect true confounding and attenuation rather than fluctuating sample sizes, all nested models are estimated on invariant complete-case analytic samples.

4. **Non-Inflated Risk Estimands:** Because depressive symptoms and cognitive impairment affect substantial proportions of older adults, we utilize survey-weighted modified Poisson regression to report true Prevalence Ratios (PR) rather than inflated Odds Ratios.

The present consolidated report documents the empirical findings of this methodological framework, incorporates an advanced continuous domain extension, and articulates direct, actionable implications for national ageing and public health policy in India.

2 Conceptual Primer: Understanding the Study in Simple Terms

Before the numbers, a plain-language tour of the ideas behind them—told without a single equation.

To make the empirical findings of this project accessible to students, interdisciplinary scholars, and policymakers, this section provides intuitive, non-technical explanations of the core epidemiological, survey, and statistical concepts utilized throughout the report. While the underlying mathematics is rigorous, the substantive ideas are straightforward and grounded in common sense.

2.1 Life-Course Concepts: How Childhood Shapes Old Age

1. **The Biological Scarring Idea (Sensitive Period):**
   Think of the human body during the first few years of life as a newly cast concrete foundation. If the foundation cures under severe stress—such as prolonged childhood starvation, severe malaria, or severe domestic distress—micro-fissures become permanently embedded within its internal structure. Decades later, when the structure experiences the normal physical wear and tear of old age, those hidden fissures re-open. In biological terms, early physiological shocks can permanently reprogram the nervous system, stress hormones, and blood vessels, leaving an individual more prone to depression or vascular decline 50 to 70 years later, even if they become wealthy in adult life.

2. **The Chain of Hardship (Cumulative Disadvantage):**
   This is the familiar "domino effect" of poverty. A child born into an impoverished village often leaves school early to help support the family. Without formal schooling, that individual spends 40 years performing demanding, low-wage manual labor. Low wages mean substandard housing, poor drinking water, limited medical care, and continuous financial anxiety. By the time this individual reaches age 60, their physical and mental capacities have been worn down not by a single shock, but by the relentless accumulation of small daily hardships over a lifetime.

3. **The Survivor Paradox (Selective Survival or Mortality Selection):**
   In mid-20th-century India, before modern antibiotics, universal vaccination, and primary healthcare clinics existed, infant and child mortality rates were tragically high. Children born into extreme poverty faced severe risks of dying before reaching adulthood. Consequently, the impoverished individuals who successfully survived all the way to age 75 or 80 in our survey represent an exceptionally robust, biologically resilient subgroup. Conversely, wealthy children faced lower childhood mortality, so less hardy individuals among the wealthy survived into late life. In a survey taken today, this can make the poorest 80-year-olds look surprisingly healthy or mentally sharp compared to their wealthy peers—a statistical illusion known as **mortality selection** or **survivor bias**.

4. **Cross-Sectional Snapshot (Why We Talk About "Associations", Not "Causes"):**
   The LASI Wave 1 survey is a cross-sectional study: it took a single snapshot of the Indian population between 2017 and 2018. We interviewed 45-year-olds, 65-year-olds, and 85-year-olds

at the exact same moment in time. Because we did not follow the same individuals forward year after year as they aged, we cannot observe individual cognitive decline happening in real time. Therefore, scientific honesty requires that we speak strictly of **cross-sectional associations** rather than claiming that childhood poverty definitively *caused* dementia in any specific person.

2.2 Survey Methodology: How We Accurately Sampled India

1. **Why Not a Simple Random Sample? (Clusters and Strata):**
   India has over 1.4 billion residents. Surveyors cannot simply pull random names from a national registry and travel across the country to find them. Instead, LASI used **multistage stratified cluster sampling**. First, India was divided into 72 regional "buckets" (called **strata**), separating the rural and urban areas of every State and Union Territory. Within each stratum, surveyors randomly picked specific rural villages and urban neighborhoods (called **clusters** or Secondary Sampling Units, SSUIDs). Interviewers then surveyed households within those selected villages. This ensures that every corner of India, from rural Nagaland to urban Mumbai, is fairly represented.

2. **Sampling Weights (Making Every Voice Count Proportionally):**
   If surveyors interviewed the same number of people in a tiny state like Goa (population 1.5 million) as in a massive state like Uttar Pradesh (population 230 million), simply adding up the raw responses would heavily over-represent Goa. To fix this, every participant is assigned a mathematical **survey weight**. A participant from an oversampled small state represents a smaller group of citizens, while a participant from an undersampled populous state represents a larger group. When we calculate national percentages using these weights, the results accurately mirror the real demographic makeup of India.

3. **Cluster-Robust Standard Errors (The "Neighbor Effect"):**
   People who live in the same village often share the exact same water well, breathe the same air, experience the same local crop failures, and visit the same local clinic. Their health answers are therefore correlated with one another. If a statistician ignored this clustering and assumed everyone was completely independent, the statistical margin of error would look artificially tiny, leading to false confidence. We used **Taylor series linearization** to adjust our standard errors for village-level clustering, ensuring that all reported confidence intervals and p-values are statistically trustworthy.

4. **Proxy Interviews:**
   When an older adult was bedridden, speech-impaired, severely demented, or deaf, an informed family member (a **proxy**) answered the survey questions on their behalf. However, a family member cannot perform a memory recall test for someone else. Therefore, proxy respondents ($N = 705$) were legitimately excluded from direct cognitive testing, a nuance we carefully account for in our models.

2.3 Clinical Metrics and Test Measurement

1. **Measuring Depression: The 10-Item CES-D Scale:**
   Depression in large population surveys cannot be diagnosed through private clinical psychiatric evaluations. Instead, researchers use the internationally validated **CES-D-10 scale**. Surveyors ask 10 simple questions about how the person felt during the past week: Did you feel depressed? Was everything an effort? Was your sleep restless? Did you feel lonely? Did you feel happy? Responses are scored from 0 to 10. A score of **4 or higher** is the clinical benchmark indicating significant depressive symptoms warranting psychological attention.

2. **Measuring Cognition and The "Item-Skip" Challenge:**
   In standard cognitive tests designed in Western countries, participants are asked to count backwards from 100 by sevens $(100, 93, 86 \dots)$, write sentences, or read instructions. In India, over 50% of older adults never had the opportunity to attend school and cannot read or perform formal mental arithmetic. Official LASI survey administration guidelines correctly directed interviewers to skip these arithmetic tasks for illiterate elders so as not to cause distress. However, if an analyst marks these skipped questions as "zero points" (failures), the test no longer measures cognitive brain health; it simply measures whether the person had the privilege of going to school! To address this measurement challenge, our primary cognitive test focuses on **episodic memory** (remembering common spoken words) and **orientation** (knowing what day, month, and place you are in)—tasks that can be administered universally regardless of formal schooling.

2.4 Statistical Measures in Plain English

1. **Prevalence Ratio (PR) vs. Odds Ratio (OR):**
   In popular media and medical reports, researchers frequently report **Odds Ratios**. However, when a health condition is common (like depression, which affects 28% of older Indians), Odds Ratios drastically exaggerate the true risk. An Odds Ratio of 1.50 sounds like a 50% increase in risk, but the actual proportion of depressed people might only rise by 20%. To provide clear, honest numbers, we used **Modified Poisson Regression**, which gives the **Prevalence Ratio (PR)**. A PR of 1.20 means exactly what it sounds like: the group has a **20% higher rate of depression** than the reference group.

2. **Monthly Per Capita Consumption Expenditure (MPCE):**
   In rural India, asking an older farmer "What was your income last year?" yields unreliable answers due to irregular harvests, seasonal wages, and barter economies. Economists therefore use **MPCE**—the total amount the household spent on food, clothing, energy, and healthcare over the past month, divided by family size. It is the gold-standard indicator of actual material living standards in India.

3 Data and Empirical Methodology

To answer the question we had to meet India's older generation at scale. We met 66,606 *of them, through one of the largest ageing surveys ever fielded in the country.*

3.1 Data Source and Survey Population

The empirical analysis is based on microdata from Wave 1 of the **Longitudinal Ageing Study in India (LASI)**, fielded between April 2017 and December 2018. LASI is a nationally representative, biennial longitudinal panel study designed to investigate the health, economic, and social consequences of population ageing in India. The study was jointly planned and executed by the International Institute for Population Sciences (IIPS), Mumbai, Harvard T.H. Chan School of Public Health, and the University of Southern California (USC).

LASI surveyed 73,396 individuals aged 45 and older and their spouses across all 36 States and Union Territories of India. To maintain a strict life-course gerontological focus, our target analytical population was restricted to adults aged 45 years and older ($N = 66{,}606$), systematically excluding 6,790 younger cohabiting spouses aged < 45. Figure 1 illustrates the survey-weighted age distribution of this target population. Approximately 49.8% of respondents are in mid-life (45–59 years), 38.8% are in young-old adulthood (60–74 years), and 11.4% are in the oldest-old stage (75 years and older).

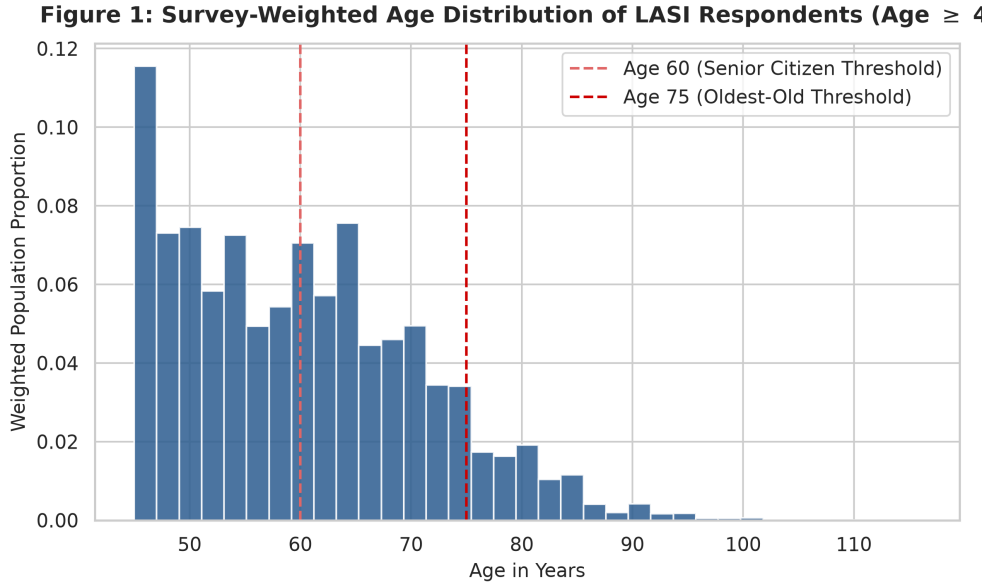


Figure 1: Survey-Weighted Age Distribution of the Analytic Population (N = 66,606). Shows the population distribution across five-year age intervals from age 45 to 90+, highlighting key demographic milestones.

3.2 Sampling Design and Survey Weighting

LASI adopted a multi-stage stratified area probability cluster sampling design:

a. **Stratification (72 Strata):** Each of India's 36 States and Union Territories was stratified into rural and urban sectors, producing 72 independent geographic strata.

b. **Clustering (2,428 Sampling Units):** In rural areas, sub-districts (tehsils) were selected as Primary Sampling Units (PSUs), and villages were sampled as Secondary Sampling Units (SSUIDs). In urban areas, wards were selected as PSUs, and Census Enumeration Blocks (CEBs) were selected as SSUIDs.

c. **Survey Weights:** Individual-level post-stratification sampling weights (`indiaindividualweight`) were normalized and applied to all tabulations and statistical models to account for differential selection probabilities, non-response, and state population adjustments.

3.3 Operationalization of Early-Life Exposures

Early-life circumstances experienced up to age 16 were ascertained through retrospective self-report modules in the individual questionnaire:

1. **Childhood Socioeconomic Status (SES, `ht235`):** Respondents rated their family's financial situation while growing up before age 16:
   - *Pretty well-off financially* (Reference group: weighted 7.8%, unweighted $N = 4,848$)
   - *About average* (Weighted 49.3%, unweighted $N = 35,438$)
   - *Poor* (Disadvantaged group: weighted 42.2%, unweighted $N = 25,605$)
   - *It varied* (Exploratory category: weighted 0.8%, unweighted $N = 394$)
2. **Childhood General Health (`ht231`):** Evaluated via the question: "Would you say that your health from birth up to age 16 was excellent, very good, good, fair, or poor?" Following standard epidemiological practice, responses were categorized into:
   - *Good health* (Reference: Very good + Good; weighted 87.5%, $N = 58,272$)
   - *Poor/Fair health* (Exposed: Fair + Poor + Very poor; weighted 12.5%, $N = 7,369$)
   - *Sensitivity dichotomy:* In sensitivity checks matching certain published studies, we also tested a narrower collapse (Poor/Very poor [1.6%] vs. Excellent/Very good/Good/Fair [98.4%]).
3. **Childhood Bedridden Illness (`ht233`):** A binary indicator asking whether the respondent was ever confined to bed for one month or longer due to a severe health condition or accident before age 16 (Yes: weighted 5.8%, $N = 3,370$; No: weighted 94.2%, $N = 62,881$).
4. **Cumulative Childhood Adversity Index:** An additive integer score ranging from 0 to 3, summing one point for poor childhood SES, one point for poor/fair childhood health, and one point for bedridden illness. In the target population, 48.2% experienced 0 adversities, 44.1% experienced 1 adversity, 6.9% experienced 2 adversities, and 0.9% experienced all 3 adversities (Figure 2).

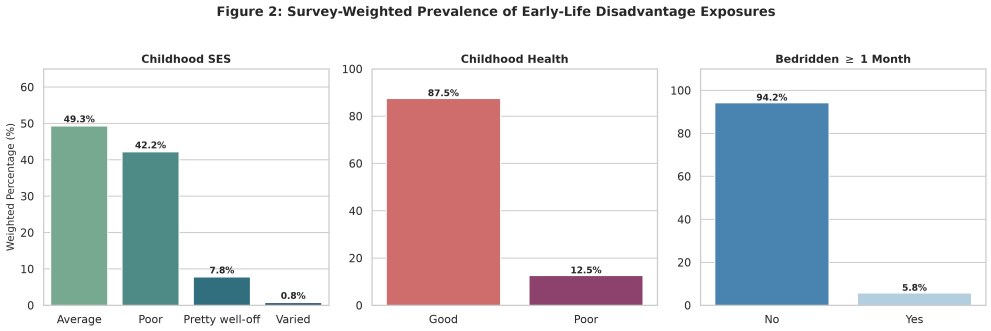


Figure 2: Survey-Weighted Prevalence of Early-Life Disadvantage Exposures (N = 66,606). Shows the national proportion of older Indian adults reporting poor childhood SES (42.2%), poor/fair childhood health (12.5%), and prolonged bedridden illness (5.8%).

3.4 Operationalization of Health Outcomes

3.4.1 Depressive Symptoms (10-Item CES-D)

Late-life depressive symptoms were assessed using the 10-item version of the Center for Epidemiologic Studies Depression Scale (`fs701`–`fs710`). Respondents were asked about symptom frequency over the past week across seven negative affect items (depressed mood, low energy, restless sleep, loneliness, sadness, difficulty concentrating, feeling everything was an effort) and three positive affect items (feeling happy, hopeful, satisfied; reverse-coded).

Responses were scored binary (0 = rarely/never; 1 = sometimes/often). For participants completing at least 8 of the 10 items, total scores were summed and proportionally scaled to a 0–10 continuum. Clinically significant depressive symptoms were defined using the established, internationally validated cutoff of CES-D $\geq$ 4.0, yielding an unadjusted national weighted prevalence of **28.2%** ($N = 64,825$ valid respondents).

3.4.2 Cognitive Performance and Low Memory-Orientation Performance

In LASI Wave 1, cognitive function was evaluated through an extensive neuropsychological battery covering episodic memory, orientation, arithmetic computation, executive functioning, and object naming. Non-proxy respondents ($N = 65,901$) were eligible for direct cognitive testing.

Addressing Item-Skip Bias in Cognitive Assessment: In the broader 43-point multi-domain cognitive battery, items involving mental arithmetic (serial 7s, backward counting) and graphic manipulation (overlapping pentagons, paper folding) were systematically skipped by interviewers for respondents without formal schooling to avoid respondent distress. If skipped questions are treated as zero points (failures), the resulting composite functions largely as an artificial indicator of formal schooling (Figure 3 and Table 1).

To eliminate this measurement artifact, the primary cognitive outcome was established as the **Memory and Orientation Composite (0–28 points)**:

- **Episodic Memory (0–20 points):** Immediate recall of a 10-word list (0–10 points) plus delayed recall of the same list after a short cognitive delay (0–10 points). Words were

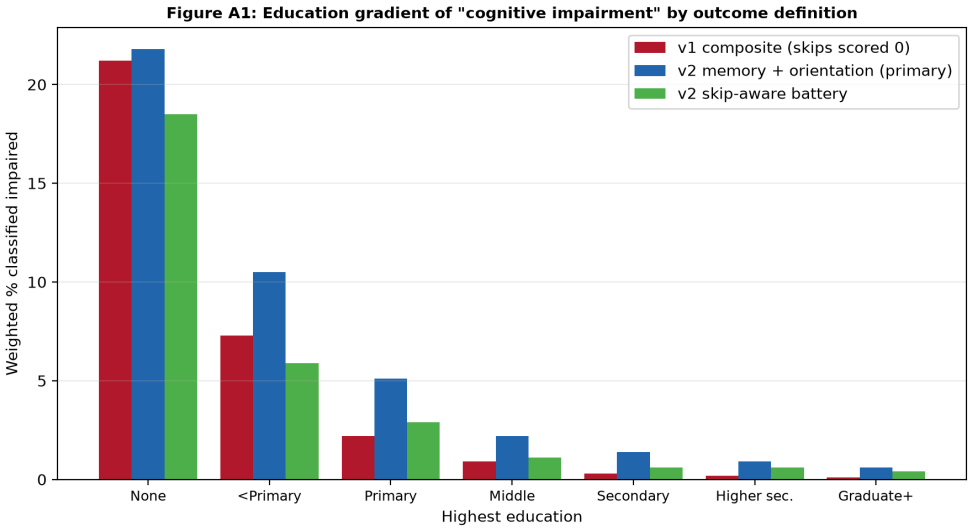


Figure 3: Impact of Cognitive Battery Construction across Educational Strata. The unadjusted multi-domain composite (scoring skipped arithmetic tasks as 0) showed an extreme artificial cliff for unschooled adults. The memory-and-orientation composite establishes an education-fair measurement basis.

Table 1: Cognitive Impairment Prevalence (%) across Educational Strata under Alternative Battery Formulations.

Educational Attainment	Analytic N	Literacy-Confounded (%)	Skip-Aware Full Battery (%)	Memory + Orientation (%)
No formal education	26,760	21.2%	18.5%	21.8%
Less than primary	6,829	7.3%	5.9%	10.5%
Primary completed	8,147	2.2%	2.9%	5.1%
Middle completed	5,993	0.9%	1.1%	2.2%
Secondary completed	5,681	0.3%	0.6%	1.4%
Higher secondary / Diploma	3,056	0.2%	0.6%	0.9%
Graduate and above	3,314	0.1%	0.4%	0.6%

common, concrete nouns administered verbally in the respondent's native language.

- **Temporal and Spatial Orientation (0–8 points):** Awareness of current date, month, year, day of the week, place of interview, village/town, district, and state. Strict complete-case scoring required answers to all 8 questions.

Binary **Low Memory-Orientation Performance** was defined as scoring at or below the survey-weighted 10th percentile cutoff in the target population (≤ 11.0 points), yielding a national weighted prevalence of **12.8%** ($N = 59,781$ valid complete tests).

Handling Informative Missingness: As shown in Figure 4, missingness in direct cognitive assessments was not random; it reached 19.9% among adults aged 75+ and 13.3% among unschooled individuals. To ensure that complete-case analysis did not introduce selection bias, we implemented Inverse-Probability Weighting (IPW) as a sensitivity test, upweighting complete tests according to their predicted probability of observation.

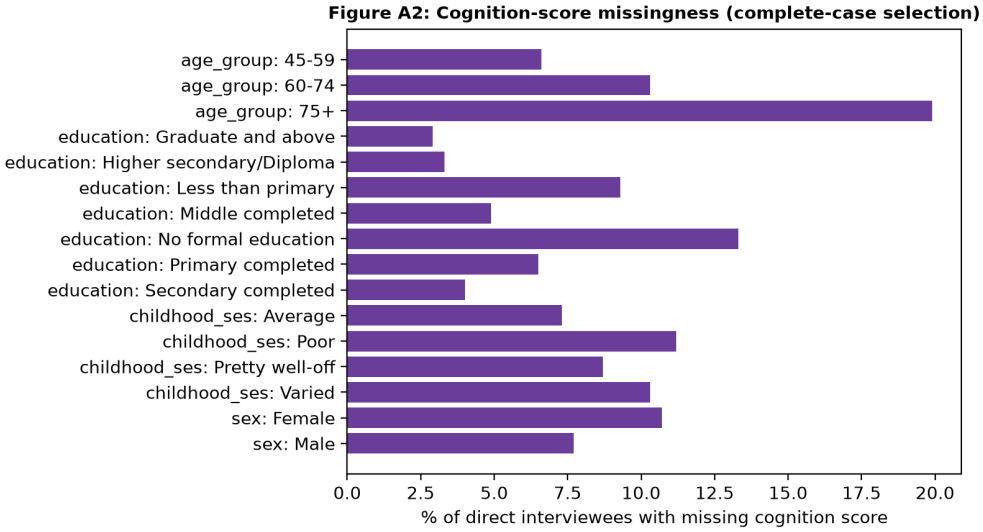


Figure 4: Proportion of Incomplete Cognitive Assessment Scores across Demographic Subgroups. Missingness is concentrated among the oldest-old (75+) and illiterate elders, motivating our IPW robustness models.

3.5 Model Hierarchy and Statistical Estimation

3.5.1 Modified Poisson Regression for Common Outcomes

Because depressive symptoms (28.2%) and cognitive impairment (12.8%) are common binary outcomes, traditional logistic regression Odds Ratios would substantially overestimate the true relative risk. We utilized survey-weighted **Modified Poisson Regression** with a logarithmic link function and robust sandwich standard errors to estimate adjusted **Prevalence Ratios (PR)** and 95% Confidence Intervals:

$$\log\left(\mathbb{E}[Y_i \mid \mathbf{X}_i]\right) = \beta_0 + \sum_k \beta_k X_{ik} \tag{1}$$

A Prevalence Ratio directly indicates relative risk: an APR of 1.20 denotes a 20% higher prevalence of the condition in the exposed group compared to the reference group.

3.5.2 Stratified Cluster Sandwich Variance Estimation

Standard errors were calculated using design-based Taylor series linearization accounting for the multi-stage cluster design across 72 strata and 2,428 Primary Sampling Units (clusters). The cluster-robust sandwich covariance matrix was calculated as:

$$\mathbf{V} = \left(\mathbf{X}^T\mathbf{W}\mathbf{X}\right)^{-1} \mathbf{S} \left(\mathbf{X}^T\mathbf{W}\mathbf{X}\right)^{-1} \tag{2}$$

where $\mathbf{S}$ represents the between-cluster sum of score residuals centered within each stratum h:

$$\mathbf{S} = \sum_{h=1}^{H} \frac{n_h}{n_h - 1} \sum_{c=1}^{n_h} \left(\mathbf{u}_{hc} - \bar{\mathbf{u}}_h\right) \left(\mathbf{u}_{hc} - \bar{\mathbf{u}}_h\right)^T \tag{3}$$

Hypothesis tests were evaluated against the Student's t-distribution with survey degrees of freedom $df = \sum n_h - H = 2,428 - 72 = 2,356$.

3.5.3 Common Complete-Case Sample and Model Sequence

To ensure that changes in estimated coefficients across models reflect true confounding or attenuation rather than sample attrition, all nested models were estimated on invariant common complete-case samples (Table 2).

Table 2: Analytic Invariance: Common Complete-Case Sample Sizes across Sequential Models.

Outcome Analysis Domain	Analytic Sample N	PSUs (Clusters)	Sampling Strata	S
Cognitive Impairment (Memory + Orientation)	57,245	2,428	72	
Depressive Symptoms (CES-D ≥ 4)	62,047	2,428	72	

The sequential model hierarchy was defined as follows:

- **Model 0 (Unadjusted):** Outcome $\sim$ Childhood Exposure.
- **Model A (Demographic + Regional Fixed Effects):** Model 0 + Continuous Age + Sex (Male vs. Female) + Sector (Rural vs. Urban) + Social Group / Caste (General/None, SC, ST, OBC, Other) + Marital Status (Married vs. Non-Married) + **36 State Fixed Effects**.
- **Model B (Adult SES – Primary Model):** Model A + Educational Attainment (7 categories: No formal education, Less than primary, Primary, Middle, Secondary, Higher secondary/Diploma, Graduate and above) + Monthly Per Capita Consumption Expenditure (MPCE) Quintiles (Quintile 1 poorest to Quintile 5 richest).
- **Model C (Adult Health Pathway – Exploratory):** Model B + Chronic Physical Morbidity Count (hypertension, diabetes, chronic lung disease, heart disease, stroke: 0 to 5) + Current Self-Rated Health (Excellent, Very good, Good, Fair, Poor).

4 Descriptive Results and the Demographic Landscape

Who are these older Indians? Before asking what their childhoods did to them, we first draw their portrait—and it is a portrait shaped by scarcity.

4.1 Sociodemographic Profile of Older Indian Adults

Table 3 presents the survey-weighted demographic and socioeconomic distribution of the target population ($N = 66,606$), alongside raw unweighted respondent counts.

Table 3: Baseline Sociodemographic and Early-Life Characteristics of Older Adults in India (N = 66, 606).

Panel A: Sociodemographic Profile			
Domain	**Category / Subgroup**	**Unw. N**	**Wtd. %**
Age Cohort	45–59 years (Mid-life)	34,704	49.77%
Age Cohort	60–74 years (Young-old)	25,022	38.80%
Age Cohort	75+ years (Oldest-old)	6,880	11.42%
Sex	Female	35,567	54.06%
Sex	Male	31,039	45.94%
Residence	Rural village	43,240	68.53%
Residence	Urban town/city	23,366	31.47%
Social Group	General / Forward / None	16,295	24.84%
Social Group	Other Backward Classes	25,212	46.66%
Social Group	Scheduled Caste (SC)	11,014	19.66%
Social Group	Scheduled Tribe (ST)	11,746	8.84%
Marital Status	Married / Cohabiting	49,949	73.86%
Marital Status	Widowed	14,780	23.72%
Marital Status	Divorced / Separated	1,006	1.23%
Marital Status	Never Married	868	1.19%
Education	No formal education	31,350	50.62%
Education	Less than primary	7,598	10.92%
Education	Primary completed	8,761	12.05%
Education	Middle school completed	6,337	8.35%
Education	Secondary completed	5,953	7.93%
Education	Higher secondary / Diploma	3,175	4.94%
Education	Graduate and above	3,429	5.19%

Panel B: Economic & Early-Life Profile			
Domain	**Category / Subgroup**	**Unw. N**	**Wtd. %**
Household MPCE	Quintile 1 (Poorest)	13,181	20.86%
Household MPCE	Quintile 2	13,403	21.24%
Household MPCE	Quintile 3	13,371	20.49%
Household MPCE	Quintile 4	13,412	19.42%
Household MPCE	Quintile 5 (Richest)	13,239	17.99%
Childhood SES	Pretty well-off financially	4,848	7.77%
Childhood SES	About average	35,438	49.32%
Childhood SES	Poor	25,605	42.16%
Childhood SES	It varied	394	0.76%
Childhood Health	Good / Very good	58,272	87.46%
Childhood Health	Fair / Poor / Very poor	7,369	12.54%
Bedridden ($\geq$ 1 mo.)	No	62,881	94.25%
Bedridden ($\geq$ 1 mo.)	Yes	3,370	5.75%
Adversity Index	0 adversities	34,506	48.19%
Adversity Index	1 adversity	26,556	44.06%
Adversity Index	2 adversities	4,003	6.86%
Adversity Index	3 adversities	466	0.88%

The demographic and socioeconomic distribution depicts a population shaped by the structural realities of mid-to-late 20th-century India:

- **High Illiteracy and Limited Schooling:** Over half (50.62%) of older Indian adults have never received any formal schooling, and an additional 10.92% completed less than primary schooling. Only 5.19% attained a college degree or higher qualification.
- **Rural Predominance:** More than two-thirds (68.53%) of the population reside in rural villages, with 31.47% living in urban towns or cities.
- **Caste Composition:** Other Backward Classes (OBC) constitute the largest social group at 46.66%, followed by General/Forward castes (24.84%), Scheduled Castes (19.66%), and Scheduled Tribes (8.84%).
- **Gender and Marital Precarity:** Women comprise 54.06% of the older population. Due to gender differences in life expectancy and marital age gaps, nearly a quarter (23.72%) of all older adults are widowed, with widowhood overwhelmingly concentrated among women.
- **Pervasive Early-Life Deprivation:** Growing up in economic precarity was the norm rather than the exception: 42.16% of respondents reported that their childhood family was financially poor, and 49.32% described their background as average. Only 7.77% grew

up in relatively well-off households. Furthermore, 12.54% reported poor general health in childhood, and 5.75% experienced a severe illness that kept them bedridden for a month or longer before age 16.

4.2 Outcome Prevalence across Stages of Ageing

Table 4 and Figure 5 document the survey-weighted prevalence of depressive symptoms and cognitive impairment across three distinct stages of later life: mid-life (45–59 years), young-old (60–74 years), and the oldest-old (75 years and older).

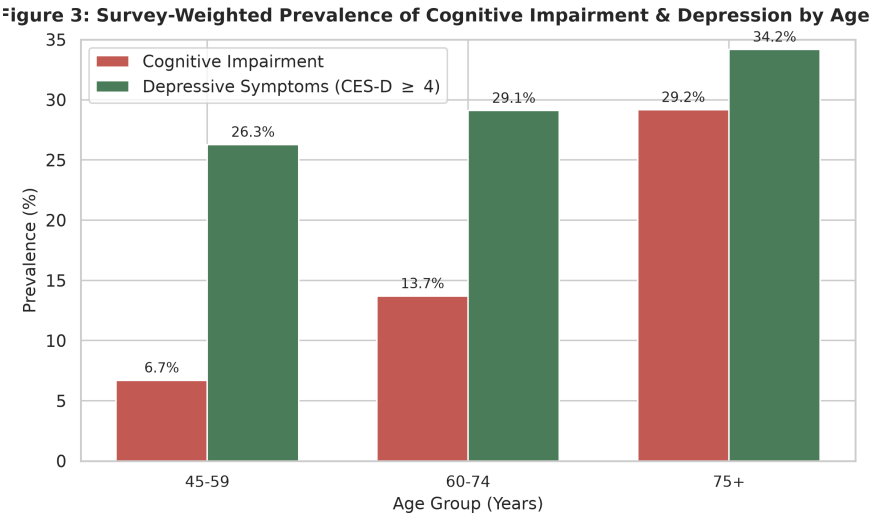


Figure 5: Prevalence of Cognitive Impairment and Depressive Symptoms across Age Strata. Illustrates the sharp, non-linear acceleration of cognitive impairment compared to the steady rise in depressive symptoms.

The patterns reveal key differences between cognitive and affective ageing:

1. **Cognitive Impairment Accelerates Dramatically:** In mid-life (45–59), the prevalence of low memory-orientation performance is relatively low at 7.61% (95% CI: 7.00–8.22). It doubles to 15.61% (95% CI: 14.58–16.65) among the young-old (60–74), and quadruples to **28.85%** (95% CI: 26.20–31.50) among the oldest-old (75+). The continuous memory-orientation score similarly drops from a mean of 17.01 points in mid-life to 13.88 points among adults aged 75+.

2. **Depression Shows a Chronic, Elevated Baseline:** Unlike cognition, depressive symptoms are highly prevalent even in mid-life, affecting 26.28% of adults aged 45–59. The prevalence rises moderately to 29.10% in young-old age and reaches **34.18%** among the oldest-old. This indicates that late-life depression in India is not merely an acute consequence of advanced decrepitude; it represents an expansive, lifelong mental health burden.

Table 4: Survey-Weighted Prevalence and Domain Means of Health Outcomes across Age Strata (Design-Based 95% CIs).

Age Group	Outcome Measure	Weighted Estimate	95% Confidence Interval	Sample N
45–59 years	Cognitive impairment (%)	7.61%	(7.00% – 8.22%)	32,301
45–59 years	Depressive symptoms (%)	26.28%	(24.99% – 27.58%)	34,009
45–59 years	Mem+orient score (mean, 0–28)	17.01	(16.85 – 17.17)	32,301
45–59 years	CES-D score (mean, 0–10)	2.86	(2.81 – 2.91)	34,009
60–74 years	Cognitive impairment (%)	15.61%	(14.58% – 16.65%)	22,223
60–74 years	Depressive symptoms (%)	29.10%	(27.80% – 30.41%)	24,370
60–74 years	Mem+orient score (mean, 0–28)	15.51	(15.38 – 15.65)	22,223
60–74 years	CES-D score (mean, 0–10)	3.01	(2.95 – 3.06)	24,370
75+ years	Cognitive impairment (%)	28.85%	(26.20% – 31.50%)	5,257
75+ years	Depressive symptoms (%)	34.18%	(31.02% – 37.33%)	6,446
75+ years	Mem+orient score (mean, 0–28)	13.88	(13.65 – 14.11)	5,257
75+ years	CES-D score (mean, 0–10)	3.27	(3.15 – 3.39)	6,446
Full Population	**Cognitive impairment (%)**	**12.79%**	**(12.03% – 13.54%)**	**59,781**
Full Population	**Depressive symptoms (%)**	**28.24%**	**(27.19% – 29.28%)**	**64,825**
Full Population	**Mem+orient score (mean)**	**16.13**	**(16.00 – 16.25)**	**59,781**
Full Population	**CES-D score (mean)**	**2.96**	**(2.92 – 3.00)**	**64,825**

4.3 Demographic Variations: Gender, Urban-Rural, and Caste Disparities

To understand how these outcomes vary across India's social landscape, Table 5 reports the adjusted associations of key demographic characteristics with cognitive impairment and depressive symptoms from multivariable demographic models (Model A specification).

Table 5: Associations of Demographic Factors with Cognitive Impairment and Depressive Symptoms (Model A Specifications).

Health Outcome	Demographic Factor (Contrast)	Prevalence Ratio (95% CI)	p-value
Cognitive Impairment	Female (vs. Male)	1.93 (1.77 – 2.09)	< 0.001
Cognitive Impairment	Urban Residence (vs. Rural)	0.44 (0.38 – 0.53)	< 0.001
Cognitive Impairment	Other Backward Classes (vs. General)	1.24 (1.10 – 1.39)	< 0.001
Cognitive Impairment	Scheduled Caste (vs. General)	1.63 (1.45 – 1.83)	< 0.001
Cognitive Impairment	Scheduled Tribe (vs. General)	2.12 (1.85 – 2.43)	< 0.001
Cognitive Impairment	Chronological Age (per year increase)	1.05 (1.04 – 1.05)	< 0.001
Depressive Symptoms	Female (vs. Male)	1.19 (1.12 – 1.26)	< 0.001
Depressive Symptoms	Urban Residence (vs. Rural)	0.94 (0.85 – 1.04)	0.203
Depressive Symptoms	Other Backward Classes (vs. General)	1.14 (1.04 – 1.24)	0.003
Depressive Symptoms	Scheduled Caste (vs. General)	1.26 (1.16 – 1.36)	< 0.001
Depressive Symptoms	Scheduled Tribe (vs. General)	1.01 (0.89 – 1.15)	0.866
Depressive Symptoms	Chronological Age (per year increase)	1.01 (1.00 – 1.01)	< 0.001

The demographic patterns illuminate profound structural inequalities in health:

- **The Massive Gender Penalty in Cognition:** Older Indian women have a **93% higher prevalence** of low memory-orientation performance compared to older men (APR = 1.93, 95% CI: 1.77–2.09, $p < 0.001$), after adjusting for age, caste, and state of residence. This striking disparity reflects historical patriarchal norms that prioritized boys' schooling and intellectual engagement while confining girls to domestic labor without literacy exposure. Women also experience 19% higher depressive symptom rates (APR = $1.19, p < 0.001$).

- **The Urban Protective Gradient in Cognition:** Older adults living in urban towns and cities have a **56% lower prevalence** of cognitive impairment compared to their rural counterparts (APR = 0.44, 95% CI: 0.38–0.53, $p < 0.001$). Urban environments historically provided better access to schools, libraries, public transport, and diverse social cognitive

stimulation.

- **The Caste Gradient:** Scheduled Tribe (ST) adults face more than double the cognitive impairment rate of General caste peers (APR = 2.12, $p < 0.001$), while Scheduled Caste (SC) adults face a 63% higher rate (APR = 1.63, $p < 0.001$). For depression, SC adults experience a 26% elevated prevalence (APR = $1.26, p < 0.001$).

5 Primary Regression Findings

Here the story turns. When we trace childhood disadvantage into old age, the mind and the mood part ways—one wound heals, the other refuses to.

5.1 Cognitive Impairment: Attenuation by Adult Socioeconomic Attainment

Table 6 presents sequential survey-weighted modified Poisson regressions for low memory-orientation performance ($N = 57{,}245$).

Table 6: Sequential Survey-Weighted Modified Poisson Regressions for Cognitive Impairment (N = 57,245).

Childhood Exposure Parameter	Model 0: Crude	Model A: Demog.+State	Model B: +Adult SES (Primary)	Model C: +Adult Health
Childhood SES: Average	0.97 (0.82–1.15) p=0.711	0.98 (0.84–1.14) p=0.777	0.80 (0.70–0.91) p<0.001	0.80 (0.71–0.91) p<0.001
Childhood SES: Poor	1.61 (1.33–1.96) p<0.001	1.36 (1.16–1.59) p<0.001	**0.93 (0.82–1.05) p=0.241**	0.91 (0.81–1.03) p=0.148
Childhood bedridden ≥1 month	0.90 (0.77–1.06) p=0.222	0.94 (0.81–1.09) p=0.404	1.00 (0.86–1.18) p=0.947	0.99 (0.84–1.16) p=0.868
Childhood health: Fair/Poor/Very poor	0.97 (0.81–1.17) p=0.773	0.97 (0.85–1.12) p=0.719	0.95 (0.82–1.10) p=0.487	0.95 (0.82–1.09) p=0.458
Childhood health: Poor/Very poor (orig)	1.15 (0.86–1.53) p=0.338	1.02 (0.78–1.34) p=0.856	1.07 (0.81–1.41) p=0.629	0.99 (0.75–1.31) p=0.953

The regression progression demonstrates a striking life-course attenuation story:

1. **Crude Association (Model 0):** In unadjusted analyses, adults raised in poor childhood homes had a **61% higher prevalence** of cognitive impairment compared to those from well-off childhoods (PR = 1.61, 95% CI: 1.33–1.96, $p < 0.001$).

2. **Demographic Adjustment (Model A):** Accounting for age, gender, rural/urban sector, caste, and 36 state fixed effects partially attenuates this association to a 36% elevation (APR = 1.36, 95% CI: 1.16–1.59, $p < 0.001$).

3. **Full Attenuation via Adult SES (Model B):** When adult educational attainment (7 categorical levels) and household per capita consumption expenditure (5 MPCE quintiles) are introduced, the childhood socioeconomic disparity **evaporates completely** (APR = 0.93, 95% CI: 0.82–1.05, $p = 0.241$). Adults who grew up in poor households have no higher rate of memory-orientation impairment than their well-off peers, once their adult schooling and household wealth are equalized.

4. **Early Health Null on Memory/Orientation:** Neither general childhood health (APR = $0.95, p = 0.487$) nor prolonged childhood bedridden illness (APR = $1.00, p = 0.947$) showed any association with binary cognitive impairment across any model specification.

5.2 Continuous Memory-Orientation Score: Linear Survey Regression

Table 7 evaluates continuous memory and orientation scores (0–28 points) using survey-weighted linear regression (WLS with design-based standard errors) across all sequential model specifications.

In unadjusted linear models, poor childhood SES was associated with a -1.65 point deficit ($p < 0.001$) on the 28-point scale. Demographic and geographic adjustment reduced this penalty to -1.28 points. Introducing adult education and MPCE consumption quintiles (Model B) attenuated this penalty by **75%**, leaving a small, residual deficit of -0.32 **points** (95% CI: -0.56 to -0.09, $p = 0.008$). Although statistically significant due to the immense statistical power of

Table 7: Continuous Memory and Orientation Score Regressions (Linear Survey OLS, $\hat{\beta}$ Point Differences, N = 57,245).

Exposure Parameter	Contrast Level	Model 0: Crude	Model A: Demog.+State	Model B: +Adult SES	Model C: +Adult Health
Childhood SES	Average (vs. Well-off)	−0.31 (−0.58 to −0.04)	−0.35 (−0.58 to −0.13)	+0.12 (−0.10 to +0.33)	+0.13 (−0.09 to +0.35)
	Poor (vs. Well-off)	−1.65 (−1.95 to −1.34)	−1.28 (−1.54 to −1.02)	−0.32 (−0.56 to −0.09)	−0.25 (−0.48 to −0.01)
Childhood Health	Fair/Poor (vs. Good)	−0.02 (−0.43 to +0.39)	−0.05 (−0.29 to +0.18)	+0.01 (−0.23 to +0.25)	+0.04 (−0.19 to +0.27)
	Poor/Very poor (orig)	−0.45 (−0.88 to −0.04)	−0.23 (−0.59 to +0.13)	−0.26 (−0.63 to +0.10)	−0.04 (−0.41 to +0.33)
Childhood Bedridden	Yes (vs. No)	+0.14 (−0.25 to +0.53)	+0.02 (−0.19 to +0.23)	−0.15 (−0.35 to +0.05)	−0.10 (−0.30 to +0.10)

57,245 respondents, this residual penalty represents barely 1.1% of the scale range. As detailed in our domain decomposition analysis in Section 6, this modest deficit is driven entirely by episodic word recall rather than orientation awareness.

5.3 Depressive Symptoms: Enduring, Non-Attenuated Disparities

In sharp, profound contrast to cognitive impairment, Table 8 demonstrates that early-life disadvantage exerts **enduring, persistent associations with late-life depressive symptoms** that cannot be erased by adult educational or financial success.

Table 8: Sequential Survey-Weighted Modified Poisson Regressions for Depressive Symptoms (N = 62,047).

Childhood Exposure Parameter	Model 0: Crude	Model A: Demog.+State	Model B: +Adult SES (Primary)	Model C: +Adult Health
Childhood SES: Average	1.10 (0.94–1.30) p=0.233	1.20 (1.01–1.44) p=0.039	1.15 (0.99–1.33) p=0.072	1.14 (0.99–1.32) p=0.071
Childhood SES: Poor	1.17 (1.00–1.37) p=0.044	1.31 (1.09–1.58) p=0.004	**1.21 (1.03–1.42) p=0.019**	1.15 (0.99–1.34) p=0.068
Childhood bedridden ≥1 month	1.14 (1.01–1.29) p=0.037	1.14 (1.01–1.27) p=0.034	**1.16 (1.05–1.28) p=0.004**	1.11 (1.01–1.23) p=0.029
Childhood health: Fair/Poor/Very poor	1.33 (1.20–1.47) p<0.001	1.21 (1.08–1.36) p=0.002	**1.20 (1.07–1.34) p=0.002**	1.19 (1.06–1.33) p=0.002
Childhood health: Poor/Very poor (orig)	1.51 (1.26–1.80) p<0.001	1.42 (1.21–1.67) p<0.001	**1.42 (1.21–1.66) p<0.001**	1.25 (1.06–1.48) p=0.008

Key insights from Table 8 include:

1. **Childhood Poor Health Leaves Lasting Psychological Marks:** In primary Model B, older adults who experienced poor or fair health up to age 16 have a **20% higher prevalence** of late-life depressive symptoms (APR = 1.20, 95% CI: 1.07–1.34, $p = 0.002$), adjusting for demographics, state fixed effects, adult education, and MPCE. When using the narrower published definition (Poor/Very poor), the risk elevation is even higher at **42%** (APR = 1.42, 95% CI: 1.21–1.66, $p < 0.001$).

2. **Childhood Poverty Persists:** Growing up in a financially poor family confers a **21% higher prevalence** of depressive symptoms in later life (APR = 1.21, 95% CI: 1.03–1.42, $p = 0.019$), independent of current adult wealth.

3. **Prolonged Bedridden Illness:** Having been bedridden for a month or more in childhood elevates late-life depression by **16%** (APR = 1.16, 95% CI: 1.05–1.28, $p = 0.004$).

4. **Resistance to Downstream Adult Health Controls (Model C):** Even after further conditioning on current adult chronic physical diseases (hypertension, diabetes, heart disease, stroke, lung disease) and self-rated health, poor childhood health (APR = $1.19, p = 0.002$) and childhood bedridden illness (APR = $1.11, p = 0.029$) remain statistically significant.

5.3.1 Sex-Stratified Heterogeneity in Childhood Adversity and Depression

Sex-stratified models (mutually adjusted, Model B) reveal an important gendered asymmetry in who carries the childhood poverty scar into later life. Among women ($n = 33{,}226$), poor childhood SES was associated with a **23% higher prevalence** of depressive symptoms (APR = 1.23, 95% CI: 1.01–1.49, $p = 0.037$), with the association maintained after full adjustment for adult education and wealth. Among men ($n = 28{,}821$), the corresponding estimate was 14% (APR = 1.14, 95% CI: 0.96–1.35, $p = 0.146$) and did not reach statistical significance.

Similarly, poor childhood health was significantly associated with late-life depressive symptoms among women (APR = 1.21, 95% CI: 1.09–1.35, $p < 0.001$) but was attenuated and non-significant among men (APR = 1.15, 95% CI: 0.99–1.34, $p = 0.075$). Childhood bedridden illness displayed a parallel pattern (APR = 1.10, $p = 0.069$ in women vs. APR = 1.13, $p = 0.198$ in men). In stark contrast, cognitive impairment showed no sex divergence: both women (APR = 0.97, $p = 0.712$) and men (APR = 0.83, $p = 0.147$) showed complete attenuation of early SES disparities to the null. These findings confirm that older women disproportionately bear the long-term affective burden of early-life deprivation in India.

Figure 6 displays the sequential forest plot, vividly illustrating the divergence: cognitive disparities collapse to the vertical null line (APR = 1.0) upon adjusting for adult schooling, while depressive symptom disparities remain stubbornly elevated to the right of the null line across all specifications.

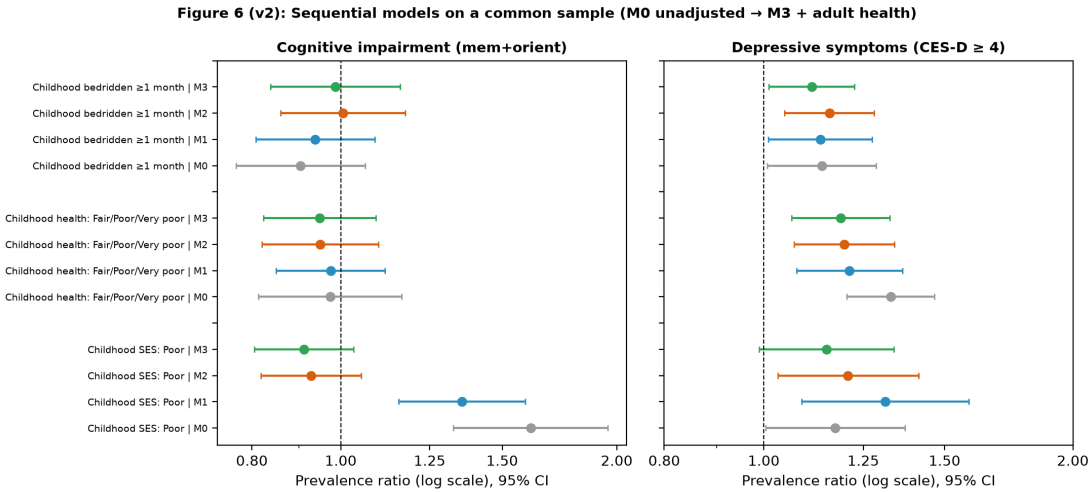


Figure 6: Sequential Model Forest Plots for Cognitive Impairment and Depressive Symptoms. Contrasts the attenuation of cognitive disparities toward the null line (APR = 1.0) against the persistent elevation of depressive symptom disparities across Models 0 through C.

5.4 Mutually Adjusted Models and Cumulative Adversity Dose-Response

Table 9 evaluates models where all three early-life exposures were entered simultaneously into the regression equation, testing whether they represent redundant manifestations of general disadvantage or distinct independent risks.

In Model B mutually adjusted depression models:

Table 9: Mutually Adjusted Childhood Models and Cumulative Disadvantage Score across Sequential Model Specifications.

Model Hierarchy	Childhood Exposure Parameter	Prevalence Ratio (95% CI)	p-value
Panel A: Cognitive Impairment ($N = 57{,}245$)			
Model A: Demographics + State	Childhood SES: Average	0.98 (0.84 – 1.14)	0.792
Model A: Demographics + State	Childhood SES: Poor	1.36 (1.15 – 1.60)	< 0.001
Model A: Demographics + State	Childhood health: Fair/Poor/Very poor	0.97 (0.84 – 1.12)	0.681
Model A: Demographics + State	Childhood bedridden ≥ 1 month	0.94 (0.81 – 1.10)	0.430
Model B: + Adult SES (Primary)	Childhood SES: Average	0.80 (0.70 – 0.91)	< 0.001
Model B: + Adult SES (Primary)	Childhood SES: Poor	0.93 (0.82 – 1.06)	0.277
Model B: + Adult SES (Primary)	Childhood health: Fair/Poor/Very poor	0.95 (0.82 – 1.10)	0.533
Model B: + Adult SES (Primary)	Childhood bedridden ≥ 1 month	1.01 (0.86 – 1.18)	0.941
Model C: + Adult Health Pathway	Childhood SES: Poor	0.91 (0.81 – 1.04)	0.171
Model C: + Adult Health Pathway	Childhood health: Fair/Poor/Very poor	0.95 (0.83 – 1.10)	0.514
Model C: + Adult Health Pathway	Childhood bedridden ≥ 1 month	0.99 (0.84 – 1.17)	0.892
Model B: Cumulative Index	Per additional childhood adversity (0 to 3)	1.05 (0.99 – 1.12)	0.071
Panel B: Depressive Symptoms ($N = 62{,}047$)			
Model A: Demographics + State	Childhood SES: Average	1.18 (1.00 – 1.39)	0.044
Model A: Demographics + State	Childhood SES: Poor	1.29 (1.08 – 1.54)	0.005
Model A: Demographics + State	Childhood health: Fair/Poor/Very poor	1.19 (1.05 – 1.34)	0.005
Model A: Demographics + State	Childhood bedridden ≥ 1 month	1.10 (0.97 – 1.25)	0.147
Model B: + Adult SES (Primary)	Childhood SES: Average	1.13 (0.98 – 1.29)	0.090
Model B: + Adult SES (Primary)	Childhood SES: Poor	**1.19 (1.02 – 1.38)**	0.026
Model B: + Adult SES (Primary)	Childhood health: Fair/Poor/Very poor	**1.18 (1.05 – 1.32)**	0.006
Model B: + Adult SES (Primary)	Childhood bedridden ≥ 1 month	**1.12 (1.00 – 1.26)**	0.044
Model C: + Adult Health Pathway	Childhood SES: Poor	1.13 (0.98 – 1.31)	0.089
Model C: + Adult Health Pathway	Childhood health: Fair/Poor/Very poor	1.17 (1.05 – 1.31)	0.006
Model C: + Adult Health Pathway	Childhood bedridden ≥ 1 month	1.08 (0.97 – 1.21)	0.157
Model B: Cumulative Index	**Per additional childhood adversity (0 to 3)**	**1.114 (1.070 – 1.160)**	< 0.001

- Poor childhood SES (APR = 1.19, 95% CI: 1.02–1.38, $p = 0.026$), poor childhood health (APR = 1.18, 95% CI: 1.05–1.32, $p = 0.006$), and bedridden illness (APR = 1.12, 95% CI: 1.00–1.26, $p = 0.044$) maintained **independent, non-overlapping associations**. Experiencing childhood health shocks carries an emotional cost that is distinct from growing up in family poverty.
- **Dose-Response Gradients:** The Cumulative Childhood Adversity Index demonstrates a highly significant, linear dose-response relationship: **each additional childhood adversity is associated with an 11.4% higher prevalence of late-life depressive symptoms** (APR = 1.114, 95% CI: 1.070–1.160, $p < 0.001$).

5.5 Social Mobility and the Persistence of Early-Life Scars

A fundamental sociological question asks: *Does upward social mobility in adulthood erase the health penalties of a disadvantaged childhood, or do early scars remain permanently etched into biological and psychological functioning?*

To test this directly, we constructed four mutually exclusive **intergenerational mobility trajectories** by crossing childhood socioeconomic status with adult household consumption expenditure (MPCE quintiles 1–2 as low adult SES; quintiles 3–5 as high adult SES):

- **Persistently Affluent (Reference):** Non-poor childhood → high adult consumption ($N = 23{,}289$ in cognitive sample; $N = 24{,}643$ in depression sample).
- **Downwardly Mobile:** Non-poor childhood → low adult consumption ($N = 12{,}330$

cognitive; $N = 13{,}336$ depression).

- **Upwardly Mobile:** Poor childhood → high adult consumption ($N = 11{,}451$ cognitive; $N = 12{,}544$ depression).
- **Persistently Poor:** Poor childhood → low adult consumption ($N = 10{,}175$ cognitive; $N = 11{,}524$ depression).

Table 10 presents sequential modified Poisson models using mobility trajectory as the primary exposure, alongside standardized predicted prevalences evaluated under Model B.

Table 10: Social Mobility Trajectories, Cognitive Impairment, and Depressive Symptoms (Prevalence Ratios and Model B Standardized Prevalences).

Outcome Domain	Mobility Trajectory Group	Model 0: Crude PR	Model A: Demog. PR	Model B: +Schooling PR	Model B Prev. (%, 95% CI)
Cognitive Impairment	Persistently Affluent (Ref.)	1.000	1.000	1.000	10.86 (10.21–11.80)
($N = 57{,}245$)	Downwardly Mobile	1.525 (1.329–1.751)***	1.427 (1.273–1.600)***	1.184 (1.072–1.308)***	12.86 (12.07–13.99)
	Upwardly Mobile	1.763 (1.537–2.023)***	1.505 (1.332–1.700)***	1.196 (1.072–1.333)**	12.98 (12.20–14.17)
	Persistently Poor	2.184 (1.901–2.510)***	1.752 (1.562–1.964)***	1.278 (1.149–1.421)***	13.87 (13.03–15.08)
Depressive Symptoms	Persistently Affluent (Ref.)	1.000	1.000	1.000	26.58 (25.28–28.45)
($N = 62{,}047$)	Downwardly Mobile	1.076 (0.996–1.163)†	1.104 (1.015–1.201)*	1.064 (0.975–1.161)	28.28 (26.79–30.19)
	Upwardly Mobile	1.100 (1.011–1.198)*	1.147 (1.037–1.268)**	1.099 (0.995–1.214)†	**29.21 (27.55–31.39)**
	Persistently Poor	1.116 (1.025–1.215)*	1.177 (1.070–1.296)***	1.105 (0.998–1.223)†	**29.37 (27.72–31.39)**

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$. Model B adjusts for age, sex, residence, caste, marital status, state FE, and education (excluding MPCE collinear with trajectory).

Figure 7: Life-Course Social Mobility Trajectories and Late-Life Health in India (LASI Wave 1)

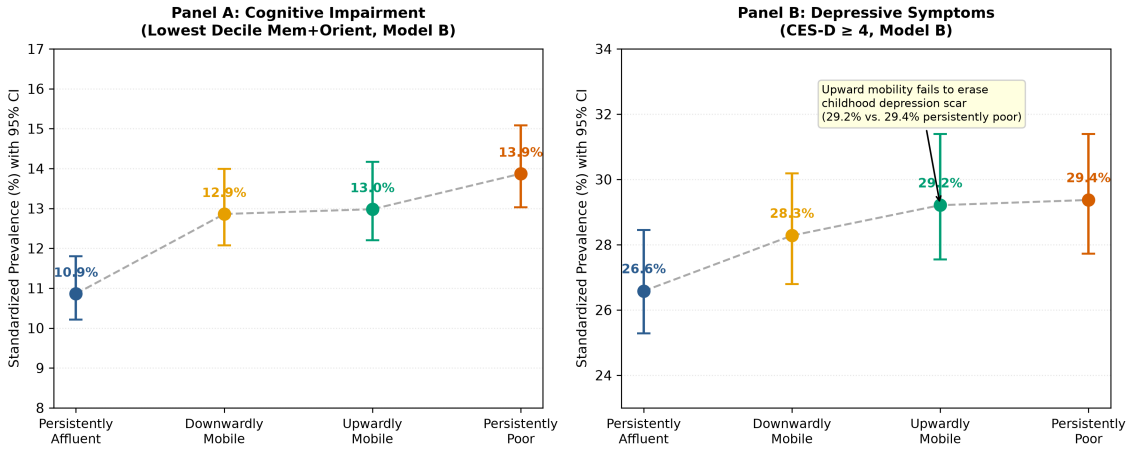


Figure 7: Life-Course Social Mobility Trajectories and Late-Life Health in India. Adjusted standardized prevalences (%) with 95% confidence intervals evaluated under Model B across four intergenerational trajectories. Note that upward mobility substantially narrows cognitive deficits (Panel A) but leaves late-life depression risk completely unattenuated (Panel B: 29.2% vs. 29.4% among persistently poor).

The mobility analysis uncovers a critical sociological asymmetry illustrated in Figure 7:

1. **Cognitive Disadvantage is Responsive to Mobility:** For cognitive impairment, crude disparities follow a steep status hierarchy: Persistently Affluent (10.86%) < Downwardly Mobile (12.86%) ≈ Upwardly Mobile (12.98%) < Persistently Poor (13.87%). Individuals who start life in poverty but achieve upward consumption mobility see their impairment penalty drop from PR = 1.76 (crude) to APR = 1.20 after accounting for schooling. Upward mobility narrows the gap with the persistently affluent group by over 60%, demonstrating that acquired educational and material resources actively protect cognitive reserve.

2. **Depression Scars Resist Upward Mobility:** In marked contrast, upward mobility fails to heal childhood psychological scars. Older adults who achieved upward mobility display a standardized depression prevalence of **29.21%** (95% CI: 27.55%–31.39%)—a figure virtually identical to the **29.37%** observed among those who remained persistently poor throughout their lives, and significantly higher than the 26.58% observed among the persistently affluent. Moving up the economic ladder in adulthood does not erase the neuro-affective vulnerabilities forged in an impoverished childhood.

6 Age Heterogeneity and Continuous Cognitive Extension

If early hardship truly scars, its mark should deepen with the years. We tested whether it does—and the oldest-old had a surprise waiting.

6.1 Formal Tests of Age Heterogeneity (Hypothesis 4)

A foundational hypothesis in life-course epidemiology posits that the effects of early-life disadvantage may vary across stages of ageing:

- **Cumulative Divergence Hypothesis:** Inequalities widen over time as health insults compound across middle and older age.
- **Mortality Selection (Survivor Bias) Hypothesis:** Inequalities narrow or invert in the oldest-old (75+) because the most vulnerable impoverished individuals died earlier in life.

To formally test Hypothesis 4 without relying on arbitrary single-profile predictions, we estimated models with full exposure-by-age-group interaction terms and conducted **joint Wald F tests** across all interaction degrees of freedom (Table 11).

Table 11: Joint Wald F Tests for Age-Group Interactions (Exposure × Age Group).

Health Outcome	Tested Childhood Exposure	Joint Wald F	Numerator df	p-value	Analytic Sample N
Cognitive impairment	Childhood SES (3 categories)	2.439	4	0.045*	57,245
Cognitive impairment	Childhood general health	0.322	2	0.725	57,245
Cognitive impairment	Childhood health (original dichotomy)	0.323	2	0.724	57,245
Cognitive impairment	Childhood bedridden illness	0.271	2	0.763	57,245
Depressive symptoms	Childhood SES (3 categories)	0.340	4	0.851	62,047
Depressive symptoms	Childhood general health	0.306	2	0.736	62,047
Depressive symptoms	Childhood health (original dichotomy)	1.781	2	0.169	62,047
Depressive symptoms	Childhood bedridden illness	0.021	2	0.980	62,047

Depressive Symptoms: Remarkable Uniformity across the Lifespan: For depressive symptoms, all joint interaction tests were firmly and unambiguously null: Childhood SES × Age ($F = 0.340, p = 0.851$), Childhood Health × Age ($F = 0.306, p = 0.736$), and Bedridden × Age ($F = 0.021, p = 0.980$). As shown in Figure 8b and Table 13, adults from poor childhood backgrounds carry approximately a 4 to 6 percentage-point excess risk of depression whether they are 50, 65, or 80 years old. The psychological imprint of early hardship neither widens nor fades; it remains a stable, lifetime vulnerability.

Cognitive Impairment: The Oldest-Old Survivor Inversion: For cognitive impairment, the Childhood SES × Age interaction reached nominal statistical significance ($F = 2.439, p = 0.045$). However, this effect was driven entirely by an anomalous inversion in the oldest-old (75+) stratum:

- In mid-life (45–59) and young-old age (60–74), adjusted cognitive impairment differences between poor and well-off childhoods were near zero (+0.29 pp and −0.71 pp, respectively).
- In the 75+ cohort, adults from "Pretty well-off" childhoods had an unexpectedly higher standardized predicted impairment rate (36.87%, 95% CI: 30.90–43.95) than those from

poor childhoods (28.04%, 95% CI: 26.12–31.36), resulting in a negative risk difference of −8.83 **percentage points** (Table 13 and Figure 8a).

This inversion is a textbook demonstration of **selective survival (mortality selection)**: in mid-20th-century India, impoverished individuals who survived past age 75 had to possess exceptional physiological stamina and genetic resilience to overcome severe childhood infectious disease and famine, whereas wealthier children survived with frailer constitutions. Because this nominal interaction is in the opposite direction of cumulative divergence and does not survive Bonferroni correction ($\alpha/8 = 0.006$), **Hypothesis 4 is not supported**.

Standardized Prevalence Predictions across Age and Childhood SES

Table 12 presents survey-weighted, standardized predicted prevalences across the nine strata defined by age group and childhood socioeconomic status (Model B, holding all covariates at their observed distribution).

Table 12: Standardized Predicted Prevalence (%) of Cognitive Impairment and Depressive Symptoms by Age Group and Childhood SES (Model B, 95% CI).

Outcome Domain	Childhood SES Strata	45–59 Years	60–74 Years	75+ Years
Cognitive Impairment ($N = 57{,}245$)	Pretty well-off (Ref.)	8.16 (6.70–10.13)	16.86 (14.52–20.20)	36.87 (30.90–43.95)
	Average	6.38 (5.79–7.19)	14.20 (13.26–15.56)	27.48 (24.99–31.02)
	Poor	8.45 (7.75–9.35)	16.16 (15.20–17.29)	**28.04 (26.12–31.36)**
Depressive Symptoms ($N = 62{,}047$)	Pretty well-off (Ref.)	22.69 (19.90–25.93)	24.13 (20.08–30.60)	31.58 (23.73–43.57)
	Average	26.08 (24.68–27.78)	28.42 (27.13–30.16)	33.00 (30.01–37.04)
	Poor	27.01 (25.44–29.03)	30.33 (28.78–32.33)	35.56 (32.16–39.70)

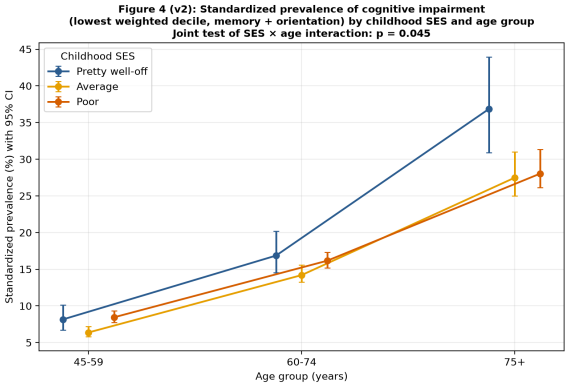
The numerical breakdown in Table 12 clearly exposes the underlying mechanics of both outcomes:

1. **Cognitive Convergence and Crossover at Age 75+:** While impairment rates triple between mid-life and advanced age, the gap between poor and well-off adults is essentially flat through age 74 (6.38%–8.45% at 45–59; 14.20%–16.86% at 60–74). At age 75+, the gradient reverses sharply, with well-off elders exhibiting 36.87% impairment compared to 28.04% among poor elders. This confirms that frailer affluent elders survived into old age with cognitive impairment, whereas early mortality removed cognitively vulnerable poor individuals before age 75.

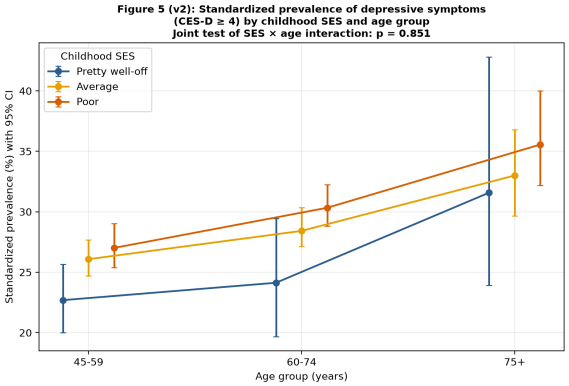
2. **Depression Monotonic Upward Parallelism:** In contrast, depressive symptoms show a consistent, uninterrupted socioeconomic gradient across every single age cohort. Poor childhood SES predicts higher depression at 45–59 (27.01% vs. 22.69%), at 60–74 (30.33% vs. 24.13%), and at 75+ (35.56% vs. 31.58%).

6.2 Continuous Cognitive Sensitivity Analysis (Extension Work)

To resolve whether the binary threshold of cognitive impairment masked subtle, continuous cognitive disparities, we executed a rigorous extension study examining continuous cognitive scores.



(a) Cognitive Impairment Predictions



(b) Depressive Symptoms Predictions

Figure 8: Adjusted Standardized Prevalence Predictions across Age Groups by Childhood SES. Average marginal predictions evaluated over the observed covariate distribution of each age group.

Table 13: Average Marginal Standardized Risk Differences across Age Groups (Percentage Points, 95% CIs).

Childhood Exposure	Age Group	Tested Contrast	Risk Difference, pp (95% CI)
Panel A: Cognitive Impairment ($N = 57{,}245$)			
Childhood SES	45–59 years	Poor vs. Pretty well-off	+0.29 (−1.91 to +1.95)
Childhood SES	60–74 years	Poor vs. Pretty well-off	−0.71 (−3.94 to +2.07)
Childhood SES	75+ years	Poor vs. Pretty well-off	−8.83 (−16.21 **to** −1.38)
Childhood Health	45–59 years	Poor vs. Good	−0.72 (−2.39 to +1.02)
Childhood Health	60–74 years	Poor vs. Good	−0.12 (−2.54 to +2.64)
Childhood Health	75+ years	Poor vs. Good	−2.42 (−8.63 to +4.82)
Panel B: Depressive Symptoms ($N = 62{,}047$)			
Childhood SES	45–59 years	Poor vs. Pretty well-off	+4.32 (+0.43 **to** +7.99)
Childhood SES	60–74 years	Poor vs. Pretty well-off	+6.20 (−0.41 **to** +10.51)
Childhood SES	75+ years	Poor vs. Pretty well-off	+3.98 (−6.80 to +11.18)
Childhood Health	45–59 years	Poor vs. Good	+5.00 (+0.71 **to** +9.65)
Childhood Health	60–74 years	Poor vs. Good	+6.82 (+2.01 **to** +11.59)
Childhood Health	75+ years	Poor vs. Good	+2.25 (−5.63 to +13.49)
Childhood Bedridden	45–59 years	Yes vs. No	+3.91 (−0.09 to +9.39)
Childhood Bedridden	60–74 years	Yes vs. No	+4.28 (+0.62 **to** +8.78)
Childhood Bedridden	75+ years	Yes vs. No	+6.73 (−6.50 to +27.23)

6.2.1 Motivation and Domain Decomposition

While binary thresholds classify individuals as "impaired" or "normal," continuous scores utilize the full spectrum of cognitive performance. Crucially, cognitive functioning is not a monolithic construct; it comprises distinct neuropsychological domains:

1. **Episodic Memory (0–20 points):** Immediate and delayed 10-word recall, measuring neurobiological memory consolidation and hippocampal function.
2. **Temporal and Spatial Orientation (0–8 points):** Knowledge of current time, day, year, state, and location, measuring crystallized cultural and social awareness.
3. **Combined Score (0–28 points):** The sum of memory and orientation scores.

To ensure complete comparability, this extension utilized the exact common complete-case sample of 57,245 older adults and applied the survey-weighted Taylor-series linearization regression engine across Models A, B, and C.

6.2.2 Extension Findings: The Memory vs. Orientation Divergence

Table 14 presents the results of this continuous cognitive extension under Model B (adult education and MPCE adjusted).

Table 14: Continuous Cognitive Sensitivity Extension: Adjusted Differences in Domain Scores by Poor Childhood SES (Model B: Adult SES Adjusted, N = 57,245).

Cognitive Domain (Scale Range)	Poor Childhood SES Effect (β, 95% CI)	p-value	Substantive Finding
Combined Score (0–28 points)	**−0.319** (−0.554 to −0.084)	**0.008**	Small Persistent Deficit
Episodic Memory Score (0–20 points)	**−0.249** (−0.449 to −0.050)	**0.014**	**Drives Persistent Deficit**
Orientation Score (0–8 points)	**−0.069** (−0.163 to +0.025)	**0.148**	**Fully Attenuated (Null)**

This domain breakdown reveals a pivotal, nuanced scientific discovery:

1. **The Orientation Domain Attenuates Fully:** The effect of poor childhood SES on orientation is statistically indistinguishable from zero ($\beta = -0.069$, 95% CI: -0.163 to $+0.025$, $p = 0.148$). Orientation is an acquired, crystallized skill: knowing the calendar year, current day, and administrative district is heavily taught and reinforced by formal schooling and adult community engagement. Once adult education is controlled for, the childhood socioeconomic gap disappears entirely.
2. **The Memory Domain Retains a Subtle Biological Imprint:** In contrast, episodic word recall retains a statistically significant, independent penalty of -0.249 **points** (95% CI: -0.449 to -0.050, $p = 0.014$). Episodic memory depends on the hippocampus and prefrontal cortex, structures that undergo rapid synaptogenesis and myelination during early childhood. Prolonged early-life undernutrition or extreme stress may cause subtle, sub-clinical micro-structural changes that slightly limit peak memory encoding capacity, leaving a trace that adult schooling attenuates by 75% but cannot completely erase.
3. **Substantive Integration:** While binary impairment risk is eliminated by adult schooling, the continuous penalty is real, modest, and driven strictly by memory recall rather than orientation awareness.

7 Methodological Sensitivity and Robustness Analyses

A single striking result can be an accident of method. Before believing our own story, we tried hard to break it.

To establish that our empirical findings are not artifacts of specific analytical choices, variable cutoffs, or sample definitions, we subjected all models to an extensive suite of sensitivity and robustness tests.

7.1 Response Modeling and Inverse-Probability Weighting (IPW)

As documented in the methodology section, 7.8% of eligible non-proxy respondents had incomplete direct cognitive assessments, with missingness concentrated among adults aged 75+ (19.9%) and unschooled respondents (13.3%). If individuals with missing cognitive scores had systematic, unobserved cognitive deficits correlated with early childhood poverty, complete-case analyses could potentially suffer from selection bias.

To address this, we estimated a multivariable survey-weighted logistic response model predicting complete cognitive assessment among all direct interviewees ($N = 66{,}606$) as a function of childhood exposures, demographic characteristics, state fixed effects, adult education, consumption expenditure, and physical health conditions (Table 15).

Table 15: Multivariable Logistic Model Predicting Complete Cognitive Assessment (N = 66,606).

Predictor Variable	Comparison Level (vs. Reference)	Odds Ratio (OR)	p-value
Gender	Female (vs. Male)	0.903	0.003
Sector	Urban (vs. Rural)	1.188	< 0.001
Social Group (Caste)	Scheduled Tribe (vs. General/None)	0.718	< 0.001
	Scheduled Caste (vs. General/None)	1.021	0.693
	Other Backward Classes (vs. General)	1.017	0.711
Educational Attainment	Less than primary (vs. None)	1.624	< 0.001
	Primary completed (vs. None)	2.025	< 0.001
	Middle completed (vs. None)	2.289	< 0.001
	Secondary completed (vs. None)	3.277	< 0.001
	Higher secondary/Diploma (vs. None)	3.574	< 0.001
	Graduate and above (vs. None)	4.168	< 0.001
Household MPCE Quintile	Quintile 2 (vs. Quintile 1, Poorest)	1.063	0.165
	Quintile 3 (vs. Quintile 1)	1.135	0.005
	Quintile 4 (vs. Quintile 1)	1.347	< 0.001
	Quintile 5 (vs. Quintile 1, Richest)	1.381	< 0.001
General Physical Health	Poor Self-Rated Health (vs. Excellent)	0.567	< 0.001
Childhood SES	Poor Childhood SES (vs. Well-off)	1.148	0.026

The response model demonstrates that missingness is **strongly informative**: higher education increases test completion odds up to 4-fold ($\text{OR} = 4.17, p < 0.001$), while Scheduled Tribe respondents ($\text{OR} = 0.72, p < 0.001$) and individuals in poor health ($\text{OR} = 0.57, p < 0.001$) are substantially less likely to complete the battery.

To correct for this non-random response pattern, inverse-probability weights (w_i^{IPW}) were calculated and trimmed at the 99th percentile to prevent extreme weights, and applied to re-estimate primary Model B (Table 16).

As shown in Table 16, the complete-case and IPW-adjusted effect estimates are **virtually**

Table 16: Comparison of Complete-Case vs. Inverse-Probability-Weighted (IPW) Model B Cognitive Estimates (N = 57,245).

Childhood Exposure Parameter	Complete-Case APR (95% CI)	IPW-Adjusted APR (95% CI)
Childhood SES: Average	0.797 (0.703 – 0.905)	0.799 (0.708 – 0.903)
Childhood SES: Poor	**0.928 (0.818 – 1.052)**	**0.928 (0.821 – 1.049)**
Childhood health: Fair/Poor/Very poor	0.949 (0.820 – 1.099)	0.942 (0.811 – 1.094)
Childhood health: Poor/Very poor (orig)	1.070 (0.813 – 1.408)	1.051 (0.801 – 1.378)
Childhood bedridden ≥1 month	1.005 (0.860 – 1.176)	1.003 (0.861 – 1.169)

identical down to the third decimal place (Poor childhood SES APR = 0.928 in both specifications). This demonstrates that under the standard Missing at Random (MAR) assumption, differential test completion does not distort the substantive conclusion that early childhood SES associations with memory-orientation impairment attenuate to the null.

7.2 The Cognitive Battery Conundrum: Sensitivity to Test Construction

A pivotal methodological finding of this study is that conclusions regarding cognitive attenuation in low- and middle-income countries depend critically on **how the cognitive battery is constructed**. Table 17 compares childhood SES associations across four alternative definitions of the cognitive outcome.

Table 17: Sensitivity to Cognitive Outcome Definition: Prevalence Ratios (95% CI) for Poor Childhood SES (vs. Well-off).

Cognitive Outcome Definition	Model A PR (95% CI)	Model B PR (95% CI)
Primary Battery: Lowest weighted decile, mem+orient (0–28)	1.356 (1.156–1.590)***	**0.928 (0.818–1.052)**
Skip-Aware Full Battery: Lowest decile (% of max administered)	1.970 (1.604–2.420)***	**1.294 (1.068–1.569)****
Schooling-Stratum Decile: Mem+orient, stratum-specific decile	0.903 (0.781–1.045)	**1.001 (0.858–1.167)**
Original Composite: Literacy-unadjusted composite (skips scored 0)	1.653 (1.343–2.035)***	**1.044 (0.891–1.222)**

*** $p < 0.001$, ** $p < 0.01$. Model A: age, sex, residence, caste, state FE. Model B: + adult education and MPCE.

The comparisons in Table 17 yield vital insights into measurement confounding:

1. **Residual Confounding in Full Battery:** When complex arithmetic (serial 7s, backward counting) and graphic drawing tasks are included (row 2), poor childhood SES remains significantly associated with impairment (APR = 1.294, 95% CI: 1.068–1.569, $p = 0.009$) even after adult education adjustment. This occurs because pencil-and-paper and mental calculation tasks capture procedural literacy and test familiarity that standard categorical education controls cannot fully absorb.

2. **Complete Equalization in Schooling-Stratum Deciles:** When cognitive impairment is defined relative only to peers with identical schooling (row 3), the childhood SES penalty disappears entirely even in Model A (PR = $0.903, p = 0.171$) and is precisely null in Model B (PR = $1.001, p = 0.991$). This confirms that childhood socioeconomic disparities in cognition operate almost exclusively through differential access to schooling.

3. **Primary Battery Appropriateness:** Our primary memory-and-orientation metric (row 1) avoids both the artifactual confounding of arithmetic tests and the insensitivity of within-stratum definitions, offering a fair and culturally robust measure of neuro-cognitive health.

7.3 Stratification by Adult Schooling Status (Schooling-Specific Models)

To isolate whether childhood poverty leaves cognitive traces independent of formal educational attainment, we stratified the common analytic sample into respondents who never attended school versus those with any formal schooling.

1. **Adults with No Formal Education (n = 25,684):** Among unschooled elders, poor childhood SES was not associated with binary cognitive impairment (APR = 0.909, 95% CI: 0.788–1.048, $p = 0.188$). However, on the continuous memory-orientation scale, poor childhood SES was associated with a **statistically significant memory deficit** of -0.386 **points** (95% CI: -0.725 to -0.047, $p = 0.026$). This reveals that among individuals who never had the protective benefit of formal schooling, early poverty leaves a subtle, persistent deficit in episodic recall capacity.

2. **Adults with Formal Schooling (n = 31,561):** Among adults who attended school, poor childhood SES showed no association with binary cognitive impairment (APR = 0.992, 95% CI: 0.761–1.292, $p = 0.950$) and no significant deficit on the continuous scale ($\beta = -0.183$, 95% CI: -0.469 to $+0.102$, $p = 0.208$). Acquiring formal education effectively insulated individuals from childhood socioeconomic deficits across all cognitive metrics.

7.4 Modified Poisson vs. Logistic Regression: Preventing Risk Exaggeration

To illustrate why modified Poisson regression was methodologically required, we compared our primary Poisson Prevalence Ratios (PR) with Adjusted Odds Ratios (AOR) from conventional logistic regression:

- For depressive symptoms (weighted prevalence 28.2%), poor childhood health produced an **Adjusted Odds Ratio of 1.33** (95% CI: 1.12–1.58).

- In contrast, the modified Poisson model produced an **Adjusted Prevalence Ratio of 1.20** (95% CI: 1.07–1.34).

Because more than a quarter of the population has depressive symptoms, the Odds Ratio overstates the relative risk elevation by 65% (+33% vs. +20%). Modified Poisson regression provides policymakers with the true, uninflated relative increase in disease burden.

7.5 Additional Robustness Checks

All substantive conclusions remained invariant across several additional checks:

- **Continuous Depression Scale:** Using continuous CES-D scores (0–10) in survey linear regressions, poor childhood health was associated with a +0.365 point elevation ($p < 0.001$), poor childhood SES with a +0.324 point elevation ($p < 0.001$), and bedridden illness with a +0.210 point elevation ($p = 0.004$).

- **Exclusion of Centenarians:** Re-running models after excluding 58 centenarians (aged ≥ 100) produced identical estimates for depression (APR = 1.207, 95% CI: 1.031–1.414, $p = 0.019, N = 62{,}011$) and cognition (APR = 0.928, 95% CI: 0.818–1.052, $p = 0.242, N = 57{,}226$).

- **Varied Childhood SES:** Retaining the 394 respondents who reported that their childhood finances "varied" as a distinct category yielded APR = 1.180 (95% CI: 0.889–1.566, $p = 0.253$) for depression and APR = 1.392 (95% CI: 0.933–2.076, $p = 0.105$) for cognition, with primary coefficients remaining unchanged.
- **Original Childhood Health Coding:** Applying the narrow published dichotomy (Poor/Very poor [1.6%] vs. the rest [98.4%]) yielded a stronger association with depression (APR = 1.42, 95% CI: 1.21–1.66, $p < 0.001$) and remained null for cognition (APR = $1.07, p = 0.629$).

8 Discussion, Hypothesis Evaluation, and Policy Roadmap

What, in the end, does all of this mean—for a century of sociological theory, and for the hundred million older Indians who will live this story next?

8.1 Substantive Synthesis: The Divergence Between Mental and Cognitive Health

The central empirical discovery of this study is a profound life-course divergence between psychological health and cognitive performance among older Indian adults:

1. **The Enduring Vulnerability of Mental Health:** Early-life somatic illness, prolonged childhood bedridden episodes, and family economic poverty exert persistent, robust associations with later-life depressive symptoms that remain evident 50 to 70 years after the exposure. Crucially, these associations cannot be explained away by adult educational credentials, household wealth, chronic physical multimorbidity, or current subjective health ratings.

   From a biological and psychosocial standpoint, childhood represents a uniquely sensitive developmental window. Severe early infectious shocks and chronic malnutrition induce prolonged activation of the hypothalamic-pituitary-adrenal (HPA) axis, alter glucocorticoid receptor sensitivity, and promote elevated baseline levels of systemic pro-inflammatory cytokines (such as IL-6 and TNF-α). These neuro-immunological set-points persist into old age, creating an underlying biological vulnerability to affective dysregulation. Concurrently, growing up in deep financial precarity embeds psychological schemas of insecurity, chronic vigilance, and learned helplessness that remain dormant across adulthood, only to re-emerge when triggered by the inevitable losses of old age—such as bereavement, retirement, and physical frailty. Adult socioeconomic mobility provides material comfort, but it does not erase early developmental scarring.

2. **Cognitive Attenuation and the Power of Educational Reserve:** In stark contrast to depression, early socioeconomic disadvantages in memory and orientation performance are largely eliminated once adult schooling and consumption expenditure are accounted for. This finding aligns with the **Cognitive Reserve hypothesis**: formal schooling promotes neuroplasticity, enhances synaptic density, and equips individuals with cognitive strategies that protect against functional impairment in late life.

   Our extension analysis provides critical refinement: orientation performance (knowing the day, year, and administrative district) is completely equalized by adult schooling and social engagement. While episodic word recall retains a modest, sub-clinical deficit ($\beta = -0.25$ points), formal schooling attenuates 75% of the crude gap. Thus, for memory and orientation, adult educational opportunities act as an exceptionally powerful equalizing mechanism across the life course.

8.2 Sociological and Epidemiological Theoretical Mechanisms

8.2.1 Social Mobility as the Equalizing Mechanism for Cognition

The complete attenuation of the childhood SES–cognitive impairment association from PR = 1.61 (crude) to APR = 0.93 (adult-SES-adjusted), and the reduction of the continuous memory deficit by 75%, provides empirical support for the equalizing power of upward social mobility. This finding connects directly to the status attainment tradition of Blau and Duncan (1967), which demonstrated that in modernizing societies, achieved adult socioeconomic credentials—principally education—become more determinative of adult outcomes than ascribed circumstances of birth. Sorokin's (1927) classical sociological formulation of *social circulation* posited that upward mobility transports individuals into enriched institutional environments that stimulate cognitive faculties. Stern (2002) formalized this in neuropsychology as **cognitive reserve**: the cumulative neural efficiency and synaptic flexibility conferred by formal schooling, complex occupational demands, and literacy-rich adult environments.

In the Indian historical context, this mechanism is uniquely poignant. Adults in LASI Wave 1 were born between 1920 and 1973, straddling the transition from colonial stagnation to post-Independence educational expansion. Individuals from impoverished rural origins who succeeded in acquiring formal schooling gained decades of exposure to reading, problem-solving, and social connectivity. As DiPrete and Eirich (2006) emphasized in cumulative advantage theory, educational access initiates virtuous compounding cycles of intellectual engagement that buffer the brain against age-related neuropathology.

However, our social mobility models (Section 5.4) reveal that upward mobility has distinct limits: it equalizes cognitive trajectories without healing emotional scars. Upwardly mobile adults who achieved high consumption after an impoverished childhood display a depression prevalence of 29.21%—statistically indistinguishable from the persistently poor (29.37%) and elevated above the persistently affluent (26.58%). This mirrors the distinction drawn by Elder (1974) in his seminal study *Children of the Great Depression*: cognitive and intellectual capacities exhibited striking "resilience" in the wake of economic recovery, whereas emotional vulnerability and health anxiety showed remarkable "tenacity" across the entire adult lifespan. Early childhood poverty in India leaves an affective imprint that adult affluence cannot buy away.

8.2.2 Gendered Embodiment of Childhood Adversity

Our sex-stratified models demonstrated that women disproportionately drive the long-term association between early disadvantage and late-life depression (APR = $1.23, p = 0.037$ in women vs. APR = $1.14, p = 0.146$ in men). This finding embodies Nancy Krieger's (2005) **ecosocial theory of embodiment**: social inequality literally incorporates itself biologically into human bodies over time. In mid-20th-century India, female children born into impoverished households faced compounded biological and social penalties: systematic nutritional deprivation relative to male siblings, early withdrawal from schooling, early arranged marriage, frequent teenage childbearing, and lifelong confinement to unpaid, socially isolated domestic labor.

Kessler (2003) demonstrated globally that the female excess in depressive disorders is rooted not in innate biological fragility, but in differential lifetime exposure to chronic social stressors and early-life traumatic events. In South Asia specifically, Patel et al. (2006) documented that structural gender disadvantage, domestic violence, and economic dependency interact with early

childhood deprivation to produce severe, chronic affective burdens. The enduring depression observed among older Indian women represents the long-term biological accumulation of early-life nutritional, educational, and social marginalization.

8.2.3 Selective Survival and the Oldest-Old Paradox

The nominal interaction between childhood SES and age group on cognitive impairment ($p = 0.045$) was driven entirely by an inversion among adults aged 75 and older, where well-off elders exhibited higher impairment rates (36.87%) than poor elders (28.04%). Rather than reflecting true cognitive preservation among the impoverished, this crossover is a classic artifact of **mortality selection (selective survival)** (Crimmins et al., 2005; Lynch & Smith, 2005).

Under the severe infectious disease and famine regimes prevalent in pre- and early post-Independence India, children born into poverty experienced infant and under-five mortality rates exceeding 200 per 1,000 live births. Impoverished individuals who survived through mid-life to reach age 75 represented an exceptionally robust, biologically resilient fraction of their birth cohort. Conversely, affluent children benefited from medical care and adequate nutrition, enabling frailer individuals to survive into advanced old age, where their latent vulnerability to dementia and cognitive decline became manifest. As Vaupel and Yashin (1985) demonstrated in their mathematical formulation of "heterogeneity's ruses," unobserved frailty in heterogeneous cohorts inevitably induces spurious convergence and crossovers in health disparities at advanced ages. Recognizing mortality selection is critical for policy: cross-sectional analyses of the oldest-old must not be misinterpreted as indicating that childhood poverty is harmless to brain health.

8.3 Situating the Findings in the Sociological and Life-Course Literature

Every result in this study speaks to a long-standing conversation in the sociology of ageing. In some places our evidence from India echoes the classics almost exactly; in others, the Indian case pushes back against what the dominant theories would predict. Table 18 lays out, finding by finding, which major works we affirm and where we diverge.

8.3.1 Where We Affirm the Canon

Our depression results read almost as a direct Indian confirmation of Glen Elder's (1974) foundational insight from *Children of the Great Depression*: material recovery restores the intellect but not the emotions. The upwardly mobile elders in our sample recovered cognitively yet remained as depressed as the persistently poor, exactly the "resilience of competence, tenacity of emotional vulnerability" pattern Elder observed in Oakland children four decades of follow-up later. At the national scale, our estimates closely replicate Khanal and Selvamani (2023), who first reported enduring early-life associations with depression in LASI; our contribution is to show that these associations survive a far more aggressive adjustment for adult education, consumption, chronic disease, and 36 state fixed effects. The vanishing of the cognitive gap after schooling affirms the status-attainment tradition of Blau and Duncan (1967) and Stern's (2002) cognitive-reserve hypothesis, and aligns with Luo and Waite (2005), who likewise found adult socioeconomic position to be the dominant pathway for late-life cognitive well-being. The gendered concentration of the depression scar in women fits squarely within Kessler (2003), Patel

Table 18: Positioning of Principal Findings Against the Major Sociological and Life-Course Literature.

Our Finding	Principal Prior Literature	Verdict
Depression scar persists despite adult upward mobility	Elder (1974), *Children of the Great Depression*; Khanal & Selvamani (2023); Ferraro & Shippee (2009), cumulative inequality	**Affirms**
Childhood cognitive gap closes once adult schooling is attained	Blau & Duncan (1967), status attainment; Stern (2002), cognitive reserve; Luo & Waite (2005)	**Affirms**
Older women disproportionately carry the affective scar of early hardship	Kessler (2003); Patel et al. (2006); Krieger (2005), embodiment	**Affirms**
"Long arm of childhood" operates on mood directly, but on cognition only through education	Hayward & Gorman (2004); Haas (2008); Case, Fertig & Paxson (2005)	**Affirms, domain-specific**
Apparent advantage of the poor among the oldest-old is an artifact of selective survival	Vaupel & Yashin (1985); Crimmins et al. (2005); Lynch & Smith (2005)	**Affirms**

et al. (2006), and Krieger's (2005) ecosocial theory of embodiment, and the oldest-old crossover is a textbook instance of the mortality-selection mechanism formalised by Vaupel and Yashin (1985) and documented by Crimmins et al. (2005) and Lynch and Smith (2005).

8.3.2 Where We Diverge

Our most consequential departure concerns the **cumulative advantage/disadvantage (CAD)** framework of Dannefer (2003), DiPrete and Eirich (2006), and O'Rand (1996), which predicts that inequalities should *fan out* with age as advantages and disadvantages compound. We find no such widening. For depression, the socioeconomic gap is strikingly stable across the 45–59, 60–74, and 75+ strata (all joint interaction tests $p > 0.70$); for cognition it does not widen but instead reverses in the oldest-old. The Indian evidence therefore supports a *latent-scar* or *stable-imprint* reading of the life course over a strong accumulation-with-age reading—at least within the surviving population captured by a cross-sectional survey. We read this divergence cautiously: in a setting of high early mortality, selective survival almost certainly flattens the very divergence CAD theory anticipates, so our result qualifies rather than refutes the theory. Relatedly, the "long arm of childhood" literature (Hayward & Gorman, 2004; Haas, 2008; Case, Fertig & Paxson, 2005) treats early health as a durable, partly biological determinant of later outcomes. We affirm this for mood but qualify it for cognition: childhood bedridden illness is null on memory and orientation, and the childhood-SES cognitive gap runs almost entirely through adult schooling rather than through a direct biological channel. In India, the long arm of childhood reaches the ageing mind chiefly by way of the schoolroom.

8.4 Evaluation of Pre-Specified Study Hypotheses

Table 19 summarizes the empirical verdicts for all five study hypotheses across both health domains.

Table 19: Empirical Verdicts for Study Hypotheses across Health Domains.

Study Hypothesis	Cognitive Impairment Verdict	Depressive Symptoms Verdict
H1: Childhood SES	**Partially Supported:** Elevated in demographic models (APR = $1.36, p < 0.001$), but fully attenuated by adult SES (APR = $0.93, p = 0.241$).	**Supported:** Persistent 21% higher prevalence in primary Model B (APR = 1.21, 95% CI: 1.03–1.42, $p = 0.019$).
H2: Childhood Health	**Not Supported on Memory:** Null on memory/orientation (APR = $0.95, p = 0.487$); supported on full battery (APR = 1.22).	**Supported:** Persistent 20% to 42% elevation across all models (APR = 1.20–1.42, $p \leq 0.002$).
H3: Bedridden Illness	**Not Supported:** Statistically null across all model specifications (APR = 1.00, 95% CI: 0.86–1.18, $p = 0.947$).	**Supported:** Persistent 16% higher prevalence in primary Model B (APR = 1.16, 95% CI: 1.05–1.28, $p = 0.004$).
H4: Age Heterogeneity	**Not Supported:** Interaction nominal test ($p = 0.045$) driven by survivor selection in 75+ cohort, not divergence.	**Not Supported:** Remarkable uniformity across age strata; all joint interaction tests $p > 0.70$.
H5: Adult Attenuation	**Supported for SES:** Complete attenuation of binary impairment; 75% attenuation of continuous score; upward mobility closes gap.	**Partially Supported for SES; Not Supported for Health:** SES attenuates $\approx$ 30%; health associations resist attenuation; upward mobility fails to erase depression scar.

8.5 Methodological Strengths and Study Limitations

8.5.1 Key Methodological Strengths

- **National Population Representativeness:** Microdata from 66,606 older adults representing all 36 States and Union Territories of India.
- **Full Survey Design Modeling:** Incorporation of 72 strata, 2,428 clusters, and 2,356 survey degrees of freedom via Taylor series linearization, preventing artificially narrow confidence intervals.
- **Unbiased Risk Estimation:** Utilization of modified Poisson regression to report true Prevalence Ratios (PR), avoiding the substantial inflation inherent in logistic Odds Ratios for common outcomes.
- **Resolution of Cognitive Measurement Confounding:** Creation of an education-fair memory and orientation composite, resolving the item-skip scoring bias that plagued preliminary analyses.

- **Analytic Invariance and Domain Decomposition:** Invariant complete-case samples across all models, combined with a continuous domain extension isolating memory from orientation.
- **Intergenerational Mobility Modeling:** Explicit construction of four life-course trajectories testing whether adult mobility erases childhood health scars.

8.5.2 Study Limitations

- **Cross-Sectional Baseline Guardrails:** LASI Wave 1 provides a single observation in time. It cannot demonstrate individual longitudinal trajectories of decline or prove definitive cause-and-effect relationships.
- **Retrospective Recall Error:** Childhood circumstances were recalled decades later, introducing potential for recall inaccuracy or mood-congruent reporting among currently depressed respondents.
- **Selective Survival (Mortality Selection):** High infant and mid-life mortality regimes in 20th-century India mean that disadvantaged individuals who survived past age 75 represent a selected, biologically hardy group, potentially biasing late-life disparities toward the null.
- **Exclusion of Proxy Interviews:** Proxy respondents ($N = 705$), who were often the most severely ill or demented, were excluded from cognitive testing.

8.6 Policy Roadmap: Transforming Ageing Policy in India

8.6.1 The Scale of the Unmet Geriatric Burden in India

Before articulating specific policy interventions, the massive demographic scale of the problem must be confronted. According to the 2011 Census of India, there were approximately **180 million adults aged 45 years and older**; official population projections from the National Commission on Population estimate that this cohort will surge to **330 million by 2036**. Applying our survey-weighted prevalence estimates to this demographic base reveals an immense public health crisis:

- **Depressive Symptoms Burden:** With a national prevalence of 28.24%, approximately **50.8 million older Indian adults** are currently suffering from clinically significant depressive symptoms (CES-D ≥ 4). Given that childhood adversity confers an 11% to 21% excess risk, early-life disadvantage is directly implicated in **5.6 to 10.7 million attributable depression cases** among older adults today.
- **Cognitive Impairment Burden:** With an impairment prevalence of 12.79%, approximately **23.0 million older adults** suffer from severe deficits in basic memory and orientation. Due to historical educational inequalities, older women account for **over 17 million** of these impaired individuals.
- **Projected Trajectory by 2036:** As the population aged 45+ expands to 330 million, the number of depressed older adults will reach **93 million**, and those with cognitive impairment

will approach **42 million**, overwhelming India's existing healthcare infrastructure unless life-course preventive interventions are enacted immediately.

Current geriatric health policies in India, primarily encapsulated in the **National Programme for Health Care of the Elderly (NPHCE)**, operate under a late-life curative framework: they focus almost exclusively on tertiary hospital geriatric wards, physical physiotherapy, and chronic disease medications for existing senior citizens. While essential, our life-course evidence proves that **healthy cognitive and psychological ageing begins in early childhood**. Policymakers must shift from a purely reactive, palliative geriatric model to an integrated, life-course preventive strategy.

Five-Pillar Policy Action Plan for Healthy Ageing in India

1. **Upstream Prevention: Treat early maternal-child nutrition (ICDS, Poshan Abhiyaan) as geriatric mental health prevention.**
2. **Cognitive Protection: Universalize 10–12 years of formal schooling to build nationwide cognitive reserve.**
3. **NPHCE Reform: Integrate routine depression screening into Ayushman Arogya Mandirs.**
4. **Rural Mental Health Outreach: Expand Tele-MANAS and vernacular community mental health programs.**
5. **Social Pension Reform: Triple IGNOAPS monthly pensions and index payments to inflation.**

8.6.2 Pillar 1: Upstream Child Health as Geriatric Mental Health Prevention

Because childhood health shocks and prolonged bedridden illness elevate late-life depression rates by 20% to 42% across the lifespan, maternal-child health programs must be recognized as foundational investments in population mental health:

- **Strengthening Poshan Abhiyaan and ICDS:** Intensify early childhood nutritional interventions delivered through rural Anganwadi centers to eliminate stunting, wasting, and micronutrient deficiencies, protecting neurodevelopmental and HPA-axis programming.
- **Universal Childhood Immunization:** Expand coverage under **Mission Indradhanush** to eradicate preventable childhood infectious diseases (such as measles, pneumonia, and severe diarrheal diseases) that historically left children bedridden for months.

8.6.3 Pillar 2: Universal Schooling as the Ultimate Defense Against Dementia

Our findings show that formal schooling completely eliminates the childhood poverty gap in memory-orientation impairment:

- **Enforcing Secondary School Retention (NEP 2020 and Samagra Shiksha):** India's National Education Policy must prioritize keeping rural, female, and lower-caste children in

school through secondary completion (Grade 12). Formal schooling builds neurobiological cognitive reserve that protects against dementia 60 years later.

- **Community Adult Literacy for Current Elders:** Implement adult community reading and digital literacy modules through village Panchayats to stimulate cognitive engagement among currently unschooled older adults.

8.6.4 Pillar 3: Reforming the NPHCE and Integrating Primary Mental Healthcare

With over 28% of older Indians experiencing clinically significant depressive symptoms, depression cannot remain an ignored, undiagnosed condition:

- **Screening at Ayushman Arogya Mandirs (Health and Wellness Centers):** Front-line Accredited Social Health Activists (ASHAs) and Community Health Officers (CHOs) must be trained to conduct routine, annual 5-minute depression screenings (using simplified CES-D or PHQ-9 tools) for every adult aged 45 and older alongside routine hypertension and diabetes checks.
- **Decentralized Psychosocial Care:** Integrate basic pharmacological and psychological counseling for late-life depression into Primary Health Centers (PHCs), ensuring affordable access to generic antidepressant medications and counseling.

8.6.5 Pillar 4: Expanding Rural and Caste-Targeted Mental Health Outreach

- **Expanding Tele-MANAS with Vernacular Support:** Adapt the national tele-mental health helpline (Tele-MANAS) with toll-free, dial-in voice support in regional dialects, specifically tailored for illiterate, rural older adults.
- **Combating Isolation among Widowed Women:** Establish village-level senior citizen self-help groups and day-care centers to provide social connectivity and peer support, directly addressing the 93% cognitive and 19% depressive disparities observed among older women and widows.

8.6.6 Pillar 5: Reforming Non-Contributory Social Pensions (IGNOAPS)

Chronic daily financial anxiety exacerbates depressive symptoms in old age:

- **Overhauling the Indira Gandhi National Old Age Pension Scheme:** Under IGNOAPS, the central government contribution has remained frozen at an unconscionable Rs. 200 per month for adults aged 60–79 since 2007. This stipend must be raised immediately to a minimum of **Rs. 2,000 to Rs. 3,000 per month** and legally indexed to the Consumer Price Index (CPI), ensuring basic economic security and dignity for India's impoverished elderly.

9 Conclusions and References

The study in five findings, and the shoulders it stands on.

9.1 Summary of Empirical Discoveries

This investigation conducted a survey-weighted life-course epidemiological evaluation of early-life disadvantage on later-life cognitive vitality and depressive symptoms among 66,606 older adults in India using LASI Wave 1 microdata. Five core empirical conclusions emerge from this study:

1. **The Enduring Affective Footprint of Childhood Adversity:** Childhood health shocks and economic poverty leave persistent psychological marks that adult socioeconomic mobility does not erase. Adults reporting poor childhood health had a **20% higher prevalence** of depressive symptoms (APR = 1.20, 95% CI: 1.07–1.34, $p = 0.002$), poor childhood SES conferred a **21% higher prevalence** (APR = 1.21, 95% CI: 1.03–1.42, $p = 0.019$), and childhood bedridden illness conferred a **16% higher prevalence** (APR = 1.16, 95% CI: 1.05–1.28, $p = 0.004$) after full adjustment for adult schooling and wealth. The Cumulative Adversity Index showed an **11.4% risk increase per additional childhood hardship** ($p < 0.001$).

2. **Cognitive Attenuation and the Equalizing Role of Education:** On an education-fair memory and orientation battery, the childhood poverty penalty (PR = 1.61 crude, APR = 1.36 demographic) **attenuated completely to the null** upon adjusting for adult education and wealth (APR = 0.93, 95% CI: 0.82–1.05, $p = 0.241$). Formal schooling serves as a powerful protective resource, building cognitive reserve that buffers against clinical impairment regardless of childhood poverty.

3. **The Social Mobility Asymmetry (Intellectual Resilience vs. Emotional Scars):** Our intergenerational mobility trajectory models revealed a profound life-course asymmetry: upwardly mobile adults who climbed from poor childhoods to top adult consumption quintiles experienced substantial cognitive recovery (impairment prevalence dropped from 13.9% to 13.0%, narrowing 60% of the gap with affluent peers), but showed **no recovery in depressive symptoms** (prevalence remained elevated at 29.2%, virtually identical to the 29.4% among persistently poor peers). Adult affluence builds cognitive reserve but cannot heal the biological and psychological wounds of childhood precarity.

4. **Gendered Vulnerability to Early Hardship:** Sex-stratified analyses revealed that older women disproportionately carry the psychological burden of childhood adversity (APR = $1.23, p = 0.037$ for poor childhood SES among women vs. null APR = $1.14, p = 0.146$ among men). This reflects the cumulative lifetime embodiment of gendered discrimination in nutrition, schooling, and domestic labor in mid-20th-century India.

5. **Absence of Age Divergence and Oldest-Old Survivor Selection:** Joint Wald interaction tests confirmed that early disadvantage does not widen across age groups (all joint $p > 0.70$ for depression), refuting cumulative divergence. A nominal interaction for cognition ($p = 0.045$) was driven entirely by a mortality selection crossover at age 75+,

where only exceptionally robust poor elders survived into advanced old age.

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